

HR / POWER

Report di identificazione

Run f0d21cfbb2404059960cc2f009dd2be1

Dataset: G2_UtD

Creazione: 2026-09-28T18:35:10.875418+00:00 | Stato: completed

Struttura: p1d_full | $\tau dx/dt + x = K (P(t-L)-P_0)$; $HR_{hat} = B+x$

Inizializzazione: estimated_equilibrium | $P_0 = \text{mean}(P_{pre})$; $HR_0 = HR(0)$; $x(0) = HR_0 - B$; B estimated; $P(t < 0) = P_0$

Dataset: 67 segmenti; train richiesti: 49; risultati: 49; validi: 49; falliti: 0.

Ambito: fit in-sample sui soli segmenti train. Val/test non utilizzati. Aggregati non pesati per durata, calcolati solo sui record fitted; n indica i valori numerici disponibili. n.d. = non disponibile o non applicabile. Nessuna conclusione fisiologica automatica.

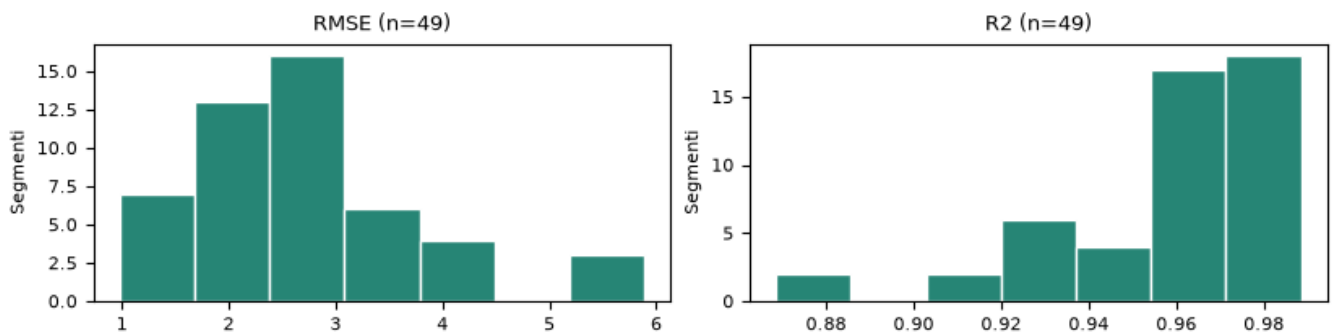
Pre-window richiesta: si; durata: 10 s. P0: pre_window_mean; HR0: first_segment_sample

Pre-window disponibili: 49 / 49 segmenti train inventariati.

Qualita della traiettoria

RMSE, MAE e bias in bpm; R2 e amplitude_ratio adimensionali. Bias positivo: HR osservata maggiore della prevista.

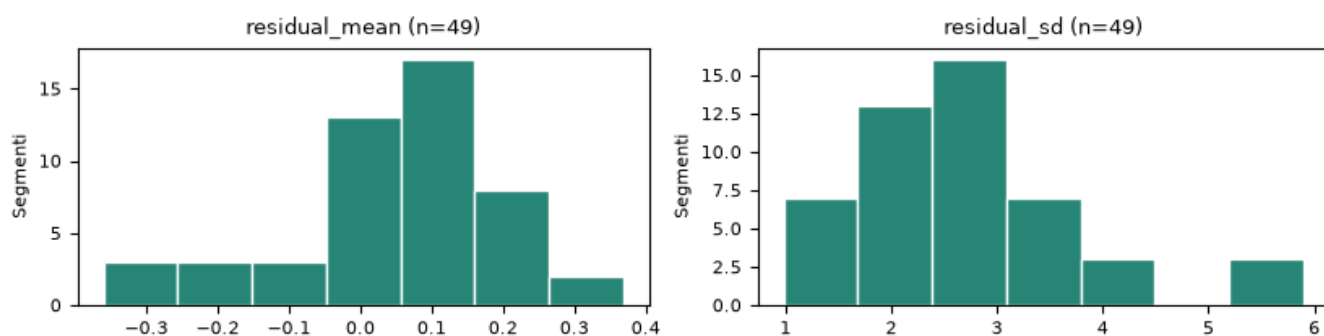
Metrica	n	Media	Mediana	SD	Min	Max
RMSE	49	2.7331	2.5902	1.1315	0.97676	5.8886
MAE	49	2.1468	2.1128	0.84165	0.77555	4.4097
bias	49	0.04988	0.072246	0.15057	-0.36025	0.36834
R2	49	0.95642	0.96108	0.026201	0.86856	0.98848
amplitude_ratio	49	1.0159	1.0129	0.074215	0.76971	1.149



Residui

e = HR osservata - HR prevista. SD campionaria (ddof=1); ACF centrata, lag in campioni. Le segnalazioni sono conservate e non impongono nuove esclusioni.

Metrica	n	Media	Mediana	SD	Min	Max
residual_mean	49	0.04988	0.072246	0.15057	-0.36025	0.36834
residual_sd	49	2.7335	2.595	1.1317	0.97903	5.8958
residual_autocorrelation_lag1	49	0.9536	0.95913	0.02977	0.83808	0.98944
residual_autocorrelation_lag5	49	0.69259	0.72185	0.17495	0.018083	0.91426



Segnalazione	Segmenti
at_bounds	1
autocorrelated_residuals	49
equivalent_predictions_different_parameters	19
multistart_disagreement	49

Multistart

Ricerca: profile_multistart. I tentativi comprendono gli eventuali raffinamenti da griglia; n_multistart indica solo gli start originali.

Start registrati: 784; convergenti: 781. Accordo di tutti gli start: 0 / 49 fit validi.

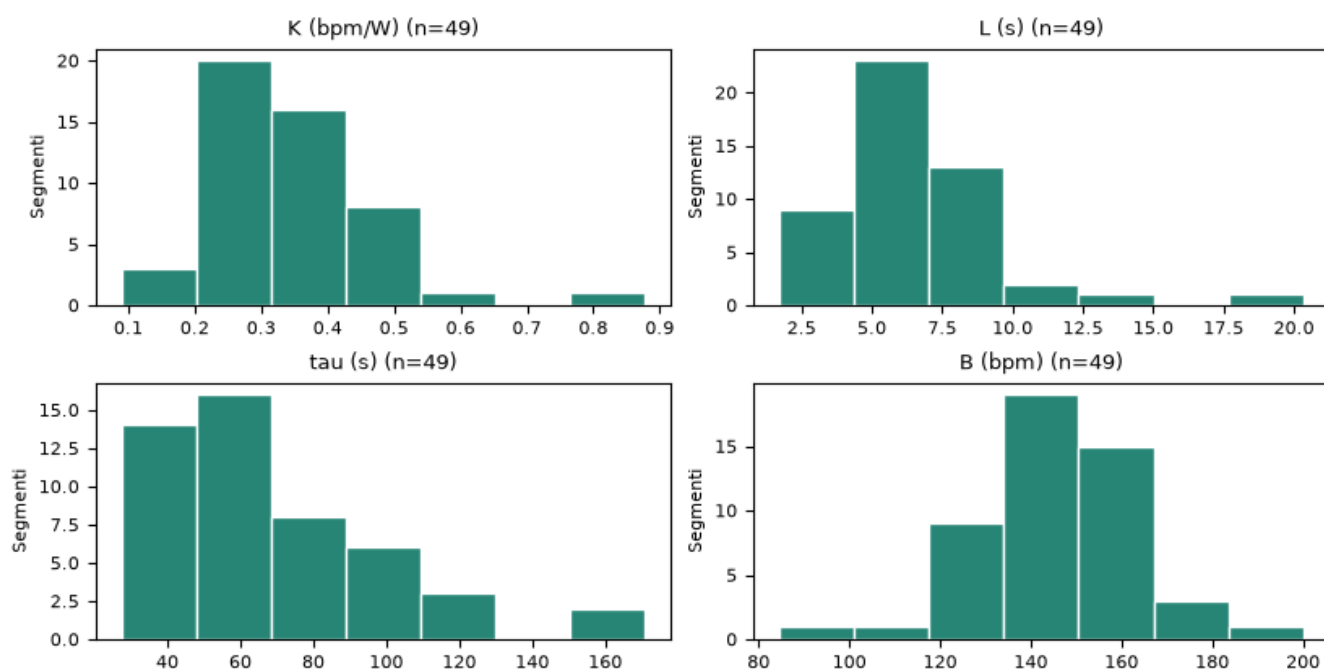
Diagnostica	n	Media	Mediana	SD	Min	Max
SSE	784	19372	2133.3	56605	192.72	3.172e+05
nfev	784	16.235	11	31.756	3	500

Distribuzione e precisione dei parametri

SE e CI95 locali di Wald, condizionati ai riferimenti iniziali. RSE in percentuale. Bounds, autocorrelazione e rango ridotto possono limitare la precisione. Un parametro fissato non ha SE/CI stimabili.

Parametro	n	Media	Mediana	SD	Min	Max
K (bpm/W)	49	0.33262	0.3176	0.12771	0.089448	0.87783
L (s)	49	6.586	6.251	3.0149	1.7117	20.322
tau (s)	49	70.08	57.377	31.707	27.38	170.87
B (bpm)	49	146.53	144.63	18.887	84.573	200

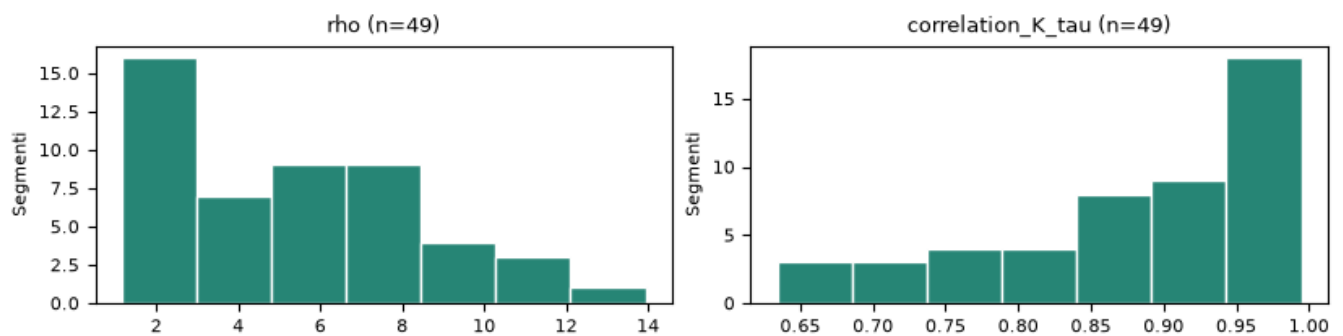
Parametro	SE mediana	RSE mediana %	AI bound	Vicino al bound
K	0.0073429	2.4677	0	0
L	0.30701	4.8783	0	0
tau	2.1326	3.7859	0	0
B	0.4394	0.29595	1	1



Identificabilita

- $\rho = (T-L)/\tau$: tempo disponibile dopo il ritardo in unita di τ . Valori piccoli descrivono una frazione breve del transitorio; nessuna soglia automatica.
- Correlazioni parametriche vicine a -1 o +1 segnalano compensazione locale tra stime; non misurano la qualita della traiettoria.
- Rango Jacobiano: confrontare con p , il numero di parametri liberi. Rango $< p$ indica direzioni non identificabili localmente.
- Multistart: SSE simili con parametri differenti indicano ambiguita. CI e correlazioni restano approssimazioni locali.

Diagnostica	n	Media	Mediana	SD	Min	Max
ρ	49	5.2946	4.909	3.1781	1.1468	13.938
correlation_K_tau	49	0.87885	0.90671	0.10258	0.63396	0.99465
Jacobian rank	49	4	4	0	4	4
Jacobian condition	49	1357.8	1016.2	933.2	402.68	4362.4



Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250625105513__laps_018-020.csv

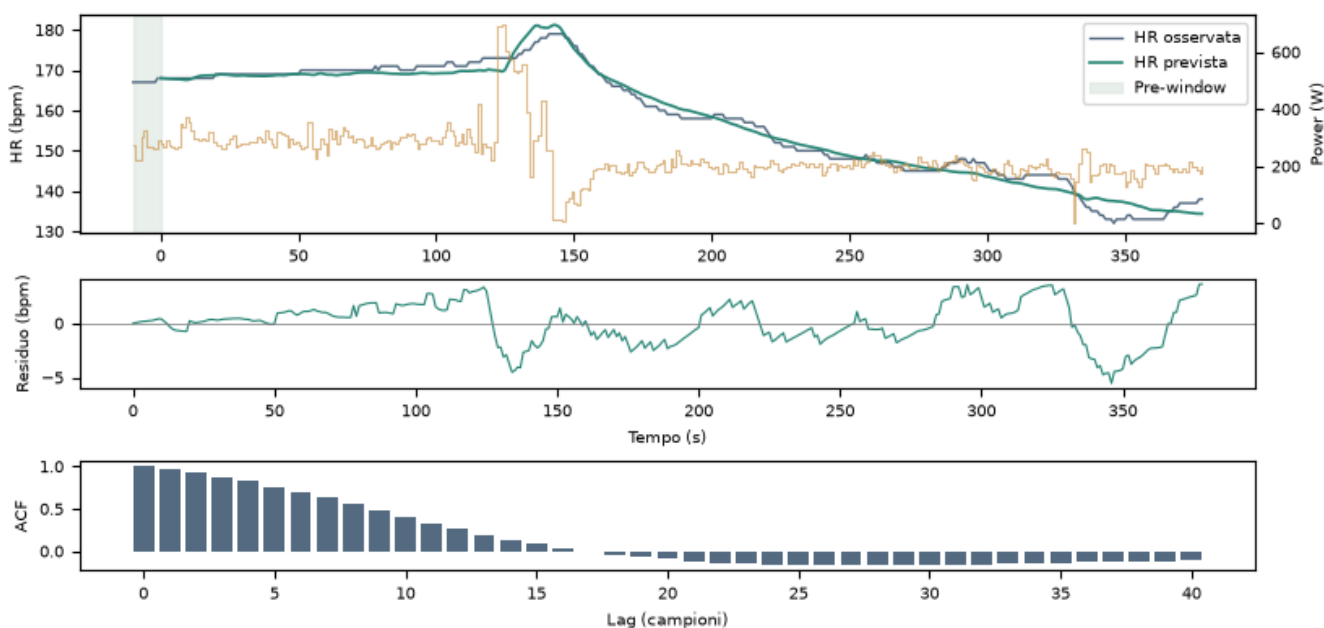
ID: 02b3e2fd2237ca893114ea2a | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=269.8 W; HR0=168 bpm.

HR(0) osservata fissata; B=160.79 bpm stimato; $x(0)=7.2141$ bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.8789	1.521	0.089793	0.98097	0.99756
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.089793	1.8792	0.964	0.76435	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.43705	0.0098547	0.41773 / 0.45636	2.2548	no	no
L	2.7263	0.36423	2.0124 / 3.4402	13.36	no	no
tau	123.43	4.8802	113.86 / 132.99	3.9539	no	no
B	160.79	0.26154	160.27 / 161.3	0.16267	no	no

rho=3.0404; Jacobian rank=4/4; cond=1531; accordo multistart=no.

Correlazioni: {"K_L": -0.21534832950661253, "K_tau": 0.9152168583787871, "K_B": -0.5530066321789514, "L_tau": -0.2602322851390675, "L_B": 0.05063049644704169, "tau_B": -0.7637654496267599}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1337.9 / 25713

Griglia: 42441 coppie L/tau; SSE multistart originale: 1337.9; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250604173939__laps_019-020.csv

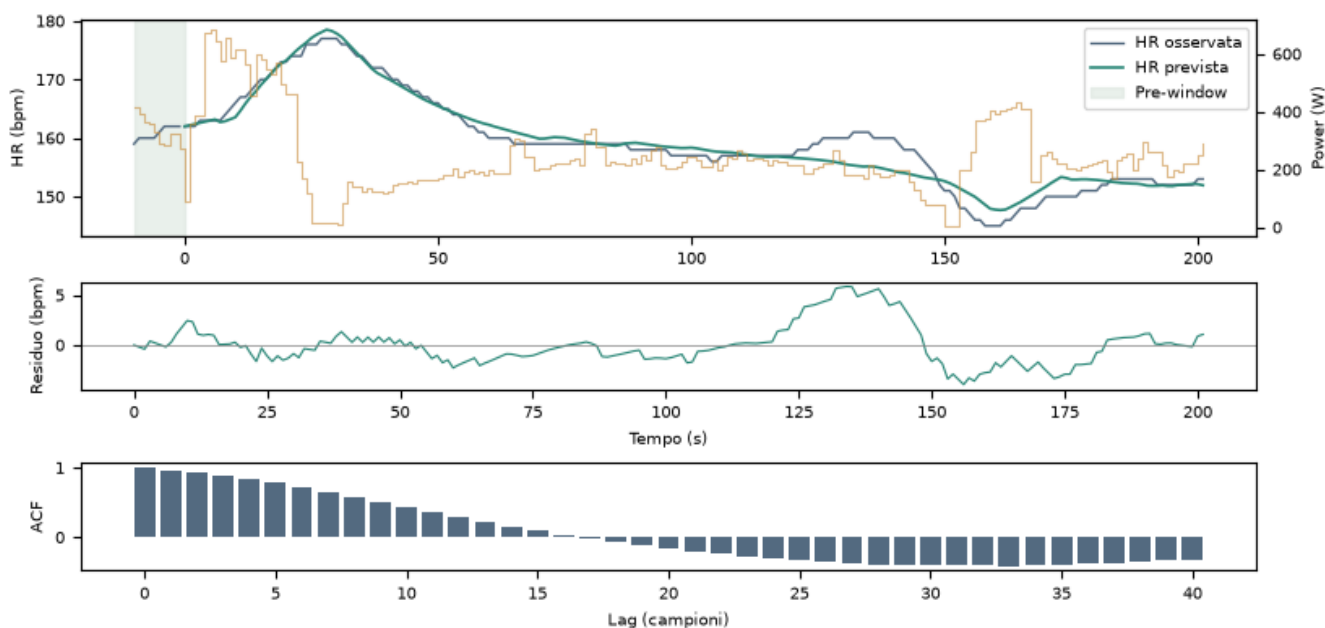
ID: 10d53032bbe7b0aee7fafa34 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=336.5 W; HR0=162 bpm.

HR(0) osservata fissata; B=182.58 bpm stimato; x(0)=-20.584 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.0218	1.4517	-0.013668	0.93212	0.96323
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.013668	2.0267	0.96743	0.78852	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.31107	0.021542	0.26885 / 0.3533	6.9249	no	no
L	6.1254	0.41325	5.3154 / 6.9354	6.7465	no	no
tau	101.18	8.5907	84.34 / 118.02	8.4907	no	no
B	182.58	1.5033	179.64 / 185.53	0.82334	no	no

rho=1.9261; Jacobian rank=4/4; cond=3635.5; accordo multistart=no.

Correlazioni: {"K_L": -0.4392110644409544, "K_tau": 0.940101717602568, "K_B": 0.9768343613896237, "L_tau": -0.4034509055947034, "L_B": -0.4826571148328546, "tau_B": 0.871877417911439}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 825.68 / 7396.4

Griglia: 42441 coppie L/tau; SSE multistart originale: 825.68; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250709173010__laps_011-012.csv

ID: 13a8b7f35941f9b2e89d2782 | Stato: fitted

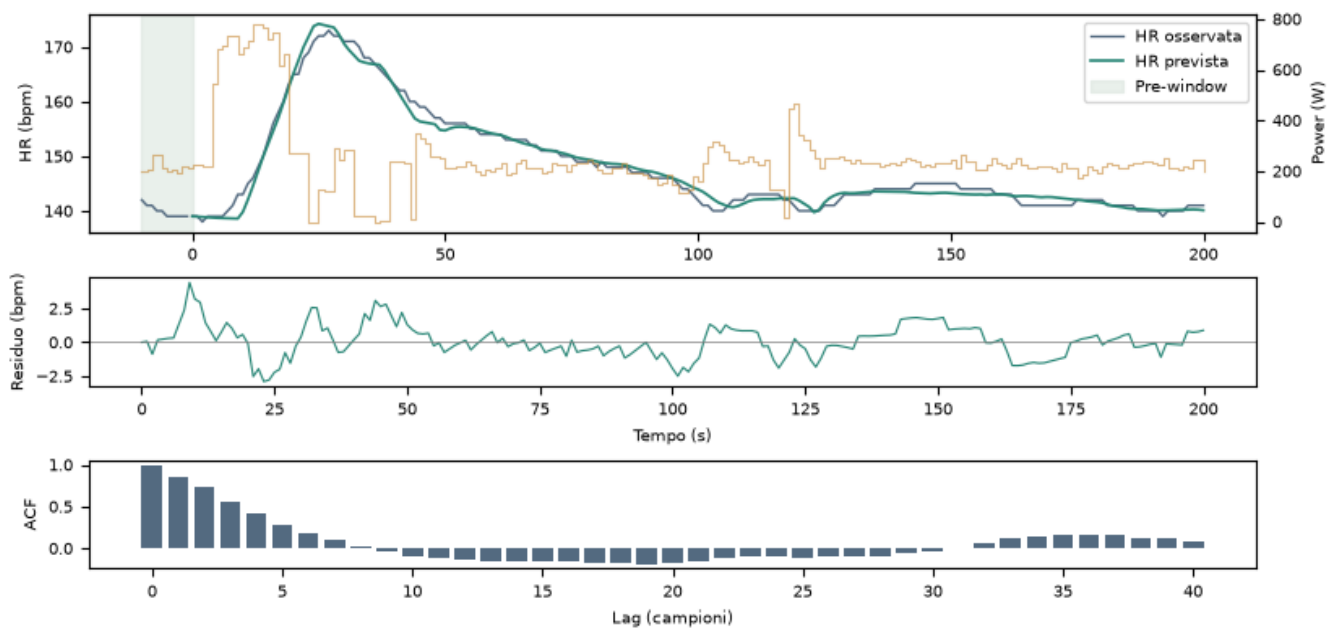
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=216.7 W; HR0=139 bpm.

HR(0) osservata fissata; B=134.51 bpm stimato; $x(0)=4.4942$ bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.1931	0.90755	0.08203	0.98255	1.0179

residual_mean	residual_sd	ACF lag 1	ACF lag 5
0.08203	1.1932	0.86236	0.28019



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.41582	0.013148	0.39005 / 0.44159	3.1619	no	no
L	5.3053	0.11741	5.0752 / 5.5355	2.2131	no	no
tau	75.259	2.7578	69.854 / 80.665	3.6644	no	no
B	134.51	0.48377	133.56 / 135.45	0.35966	no	no

rho=2.587; Jacobian rank=4/4; cond=1153; accordo multistart=no.

Correlazioni: {"K_L": -0.3653519306637097, "K_tau": 0.9690948711594203, "K_B": -0.9674719740425249, "L_tau": -0.3984720819699359, "L_B": 0.33793435140312694, "tau_B": -0.9412464696710228}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 286.1 / 15905

Griglia: 42441 coppie L/tau; SSE multistart originale: 286.1; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260321102409__laps_012-013.csv

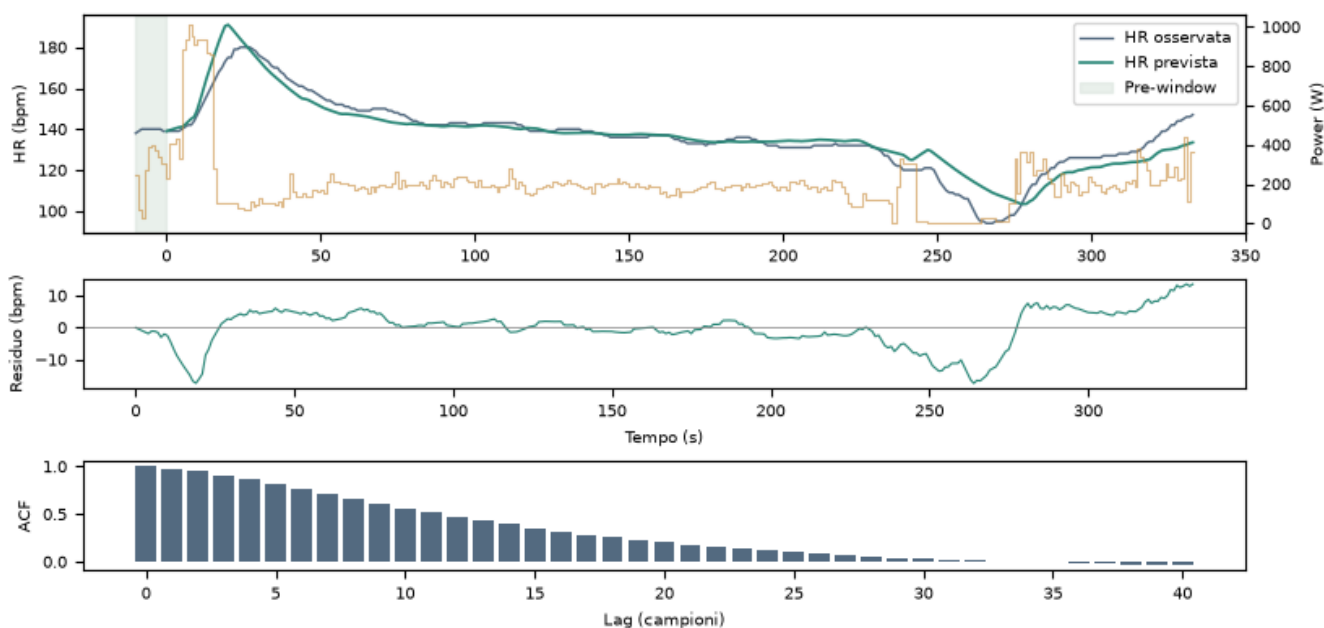
ID: 16099d4d95a84b6f5b36c5fe | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=273.4 W; HR0=139 bpm.

HR(0) osservata fissata; B=160.25 bpm stimato; x(0)=-21.246 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
5.8886	4.306	-0.13957	0.88168	1.0178
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.13957	5.8958	0.98029	0.81486	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.3176	0.010188	0.29763 / 0.33757	3.2079	no	no
L	4.305	0.35308	3.613 / 4.997	8.2016	no	no
tau	43.51	2.1676	39.262 / 47.759	4.9817	no	no
B	160.25	0.82286	158.63 / 161.86	0.5135	no	no

rho=7.5544; Jacobian rank=4/4; cond=617.86; accordo multistart=no.

Correlazioni: {"K_L": -0.34361637040371856, "K_tau": 0.755108236177321, "K_B": 0.8854275286513119, "L_tau": -0.38365443847324054, "L_B": -0.3216980769207415, "tau_B": 0.5748097074422325}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 11582 / 76073

Griglia: 42441 coppie L/tau; SSE multistart originale: 11582; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250708171459__laps_002-003.csv

ID: 1b4ade4ad920633a01b0d825 | Stato: fitted

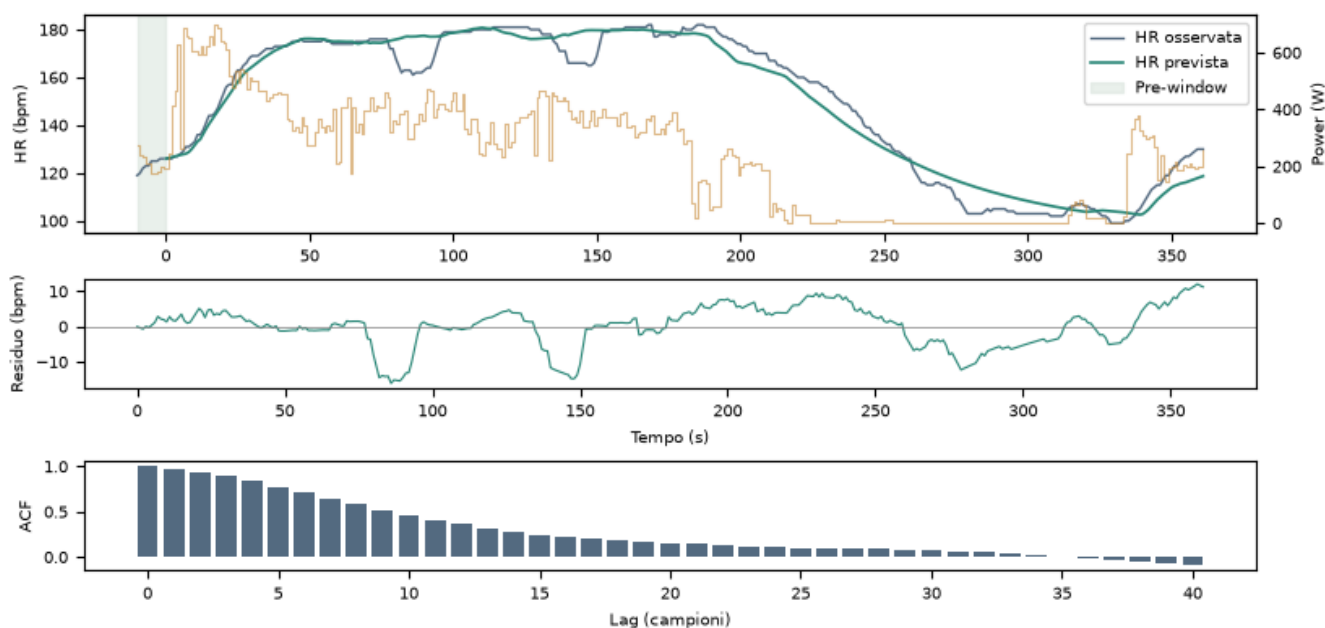
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=209.7 W; HR0=126 bpm.

HR(0) osservata fissata; B=143.16 bpm stimato; x(0)=-17.163 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
5.7862	4.4097	0.054184	0.96133	0.94225

residual_mean	residual_sd	ACF lag 1	ACF lag 5
0.054184	5.794	0.9755	0.77562



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.23444	0.0049444	0.22475 / 0.24413	2.109	no	no
L	5.6908	0.73831	4.2438 / 7.1379	12.974	no	no
tau	52.83	2.1326	48.65 / 57.01	4.0368	no	no
B	143.16	0.44579	142.29 / 144.04	0.31139	no	no

rho=6.7256; Jacobian rank=4/4; cond=968.13; accordo multistart=no.

Correlazioni: {"K_L": -0.5836233117021838, "K_tau": 0.8796160615069097, "K_B": -0.5880862450606206, "L_tau": -0.7016235289966136, "L_B": 0.23868554719073898, "tau_B": -0.5643438618655493}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 12120 / 3.0877e+05

Griglia: 42441 coppie L/tau; SSE multistart originale: 12120; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250708171459__laps_006-007.csv

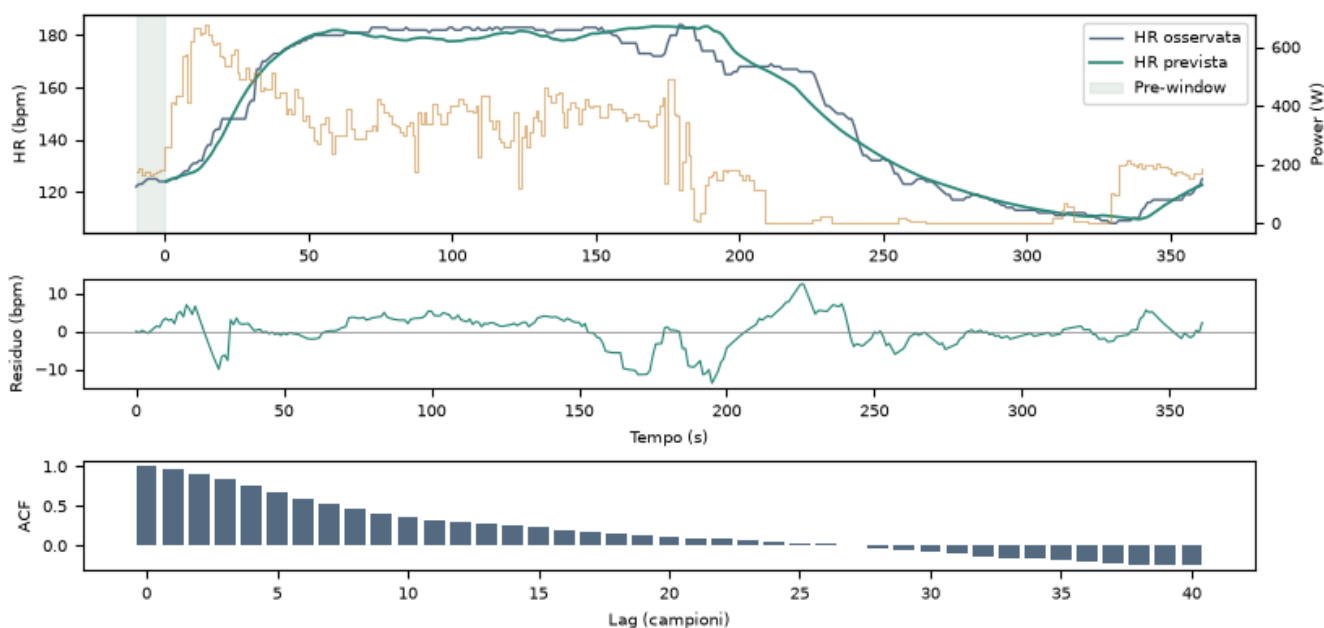
ID: 1f7a8c47b00f00f482c8b29c | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=174.4 W; HR0=124 bpm.

HR(0) osservata fissata; B=141.97 bpm stimato; x(0)=-17.967 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
4.2496	3.1519	0.10716	0.97763	0.97206
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.10716	4.2542	0.95769	0.67746	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.21631	0.0031347	0.21017 / 0.22245	1.4492	no	no
L	10.36	0.553	9.276 / 11.444	5.3379	no	no
tau	45.288	1.3286	42.684 / 47.892	2.9336	no	no
B	141.97	0.33806	141.3 / 142.63	0.23813	no	no

rho=7.7424; Jacobian rank=4/4; cond=915.34; accordo multistart=no.

Correlazioni: {"K_L": -0.5723958019340911, "K_tau": 0.8480603516583144, "K_B": -0.6543800877796445, "L_tau": -0.7252699414032607, "L_B": 0.2844205542907332, "tau_B": -0.5777650258520491}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 6537.5 / 2.8353e+05

Griglia: 42441 coppie L/tau; SSE multistart originale: 6543.6; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250708171459__laps_008-009.csv

ID: 201c1b0432475dd79dca4e7d | Stato: fitted

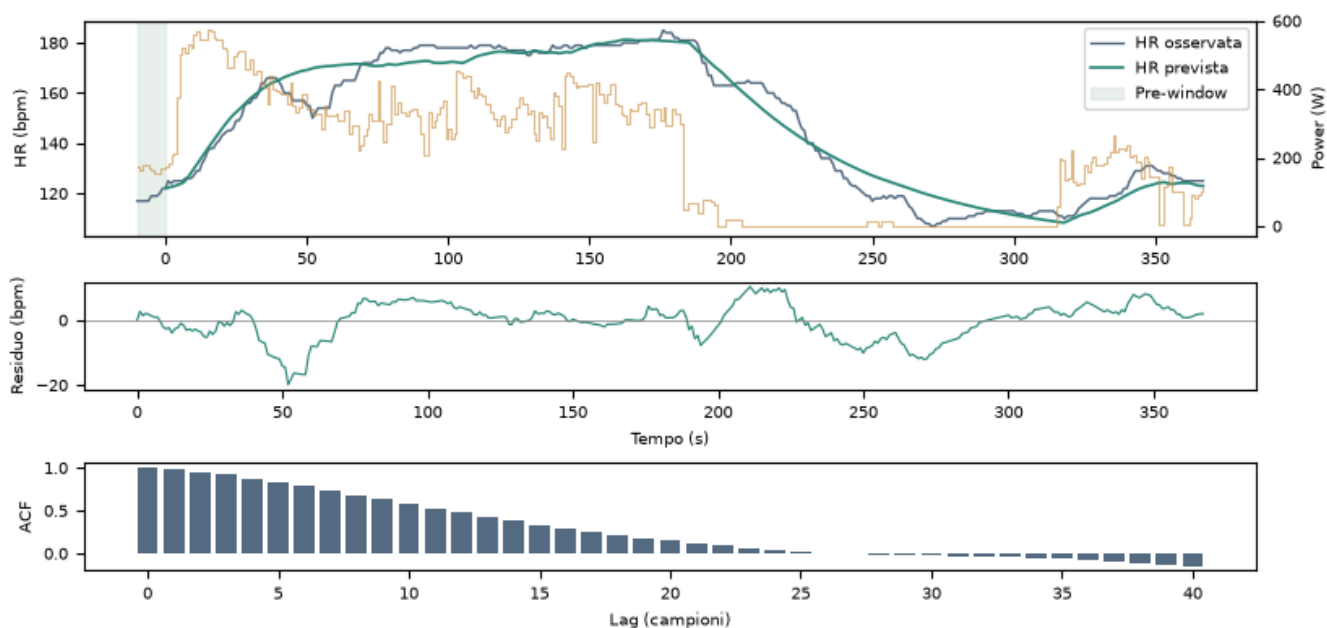
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=166.8 W; HR0=122 bpm.

HR(0) osservata fissata; B=139.38 bpm stimato; x(0)=-17.379 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
5.5477	4.2429	-0.26966	0.95693	0.93301

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.26966	5.5487	0.97838	0.83458



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.23726	0.0052478	0.22697 / 0.24754	2.2119	no	no
L	2.3506	0.80406	0.77468 / 3.9266	34.206	no	no
tau	57.377	2.2256	53.015 / 61.74	3.8789	no	no
B	139.38	0.4394	138.52 / 140.24	0.31526	no	no

rho=6.3553; Jacobian rank=4/4; cond=988.18; accordo multistart=no.

Correlazioni: {"K_L": -0.598151454680679, "K_tau": 0.8806650651921514, "K_B": -0.6271598931504827, "L_tau": -0.7197539080343114, "L_B": 0.2594208308174258, "tau_B": -0.5650782647630898}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 11326 / 2.5642e+05

Griglia: 42441 coppie L/tau; SSE multistart originale: 11326; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260919065417__laps_002-003.csv

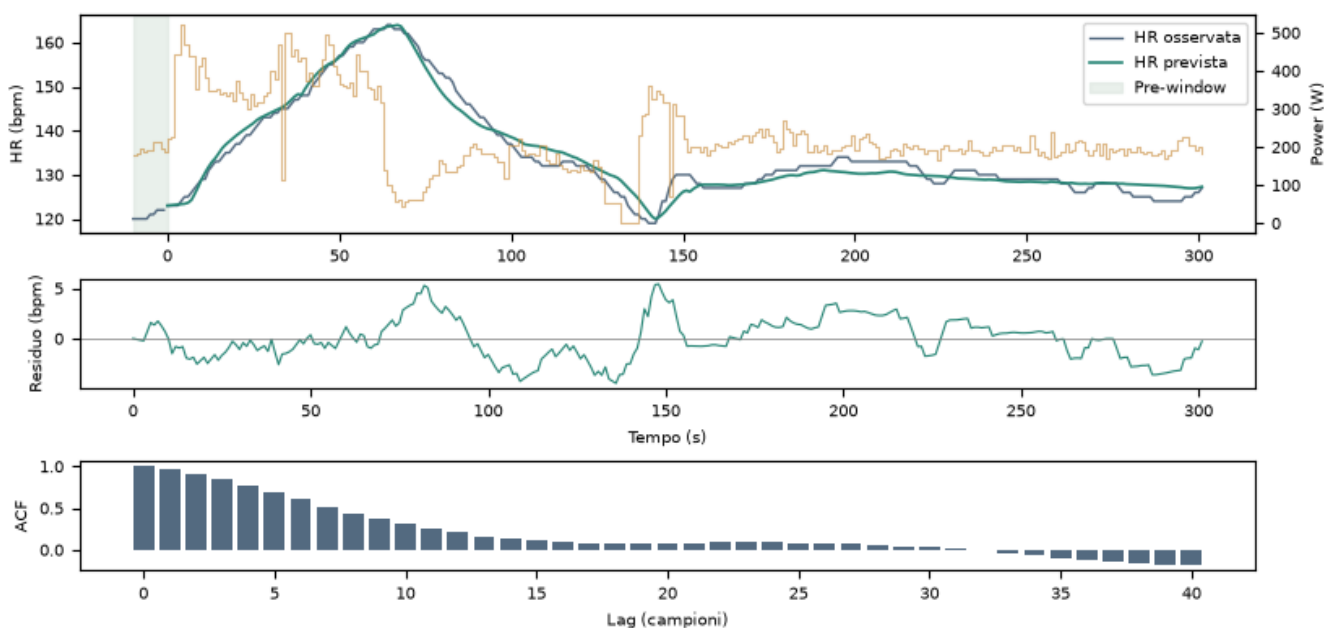
ID: 20ddf0cf07abe30440443e67 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=193.8 W; HR0=123 bpm.

HR(0) osservata fissata; B=126.91 bpm stimato; $x(0)=-3.9054$ bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.065	1.6786	-0.060244	0.96475	0.9778
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.060244	2.0676	0.96021	0.69328	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.28979	0.0063492	0.27734 / 0.30223	2.191	no	no
L	4.7012	0.27602	4.1602 / 5.2422	5.8713	no	no
tau	50.872	1.5042	47.924 / 53.821	2.9568	no	no
B	126.91	0.23201	126.45 / 127.36	0.18282	no	no

rho=5.8243; Jacobian rank=4/4; cond=616.22; accordo multistart=no.

Correlazioni: {"K_L": -0.47148756310733514, "K_tau": 0.8942477490896319, "K_B": -0.7994964199129296, "L_tau": -0.5500424677217625, "L_B": 0.3266339097947674, "tau_B": -0.7228235409133268}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1287.8 / 35584

Griglia: 42441 coppie L/tau; SSE multistart originale: 1287.8; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250625105513__laps_009-012.csv

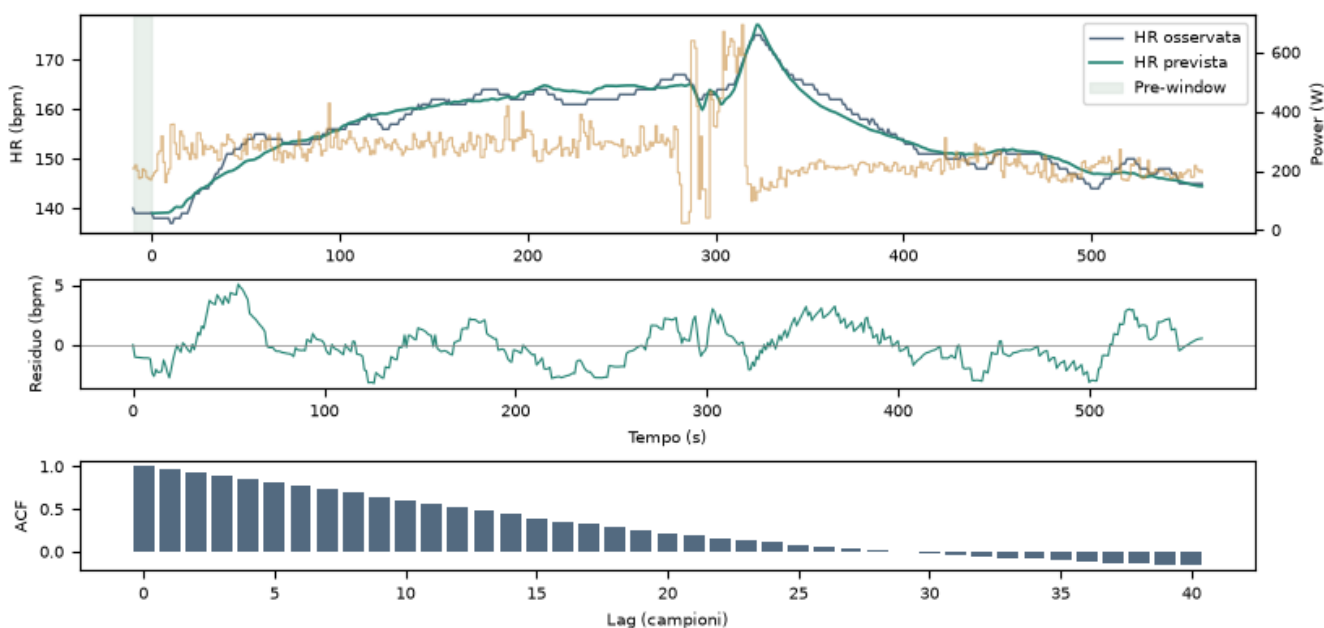
ID: 212207f51ebc4eb88edf1bc0 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=190.9 W; HR0=139 bpm.

HR(0) osservata fissata; B=140.22 bpm stimato; x(0)=-1.2247 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.6992	1.3648	0.020954	0.9558	1.0021
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.020954	1.7006	0.96314	0.81461	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.24905	0.0037335	0.24173 / 0.25637	1.4991	no	no
L	6.3128	0.25838	5.8064 / 6.8192	4.0929	no	no
tau	66.865	1.2643	64.387 / 69.343	1.8908	no	no
B	140.22	0.2743	139.69 / 140.76	0.19562	no	no

rho=8.2657; Jacobian rank=4/4; cond=1272.8; accordo multistart=no.

Correlazioni: {"K_L": -0.176229126239317, "K_tau": 0.7164696564630139, "K_B": -0.9562487871135082, "L_tau": -0.3372765260132711, "L_B": 0.1479381712290638, "tau_B": -0.6597700325911963}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1616.9 / 26964

Griglia: 42441 coppie L/tau; SSE multistart originale: 1616.9; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250709173010__laps_009-010.csv

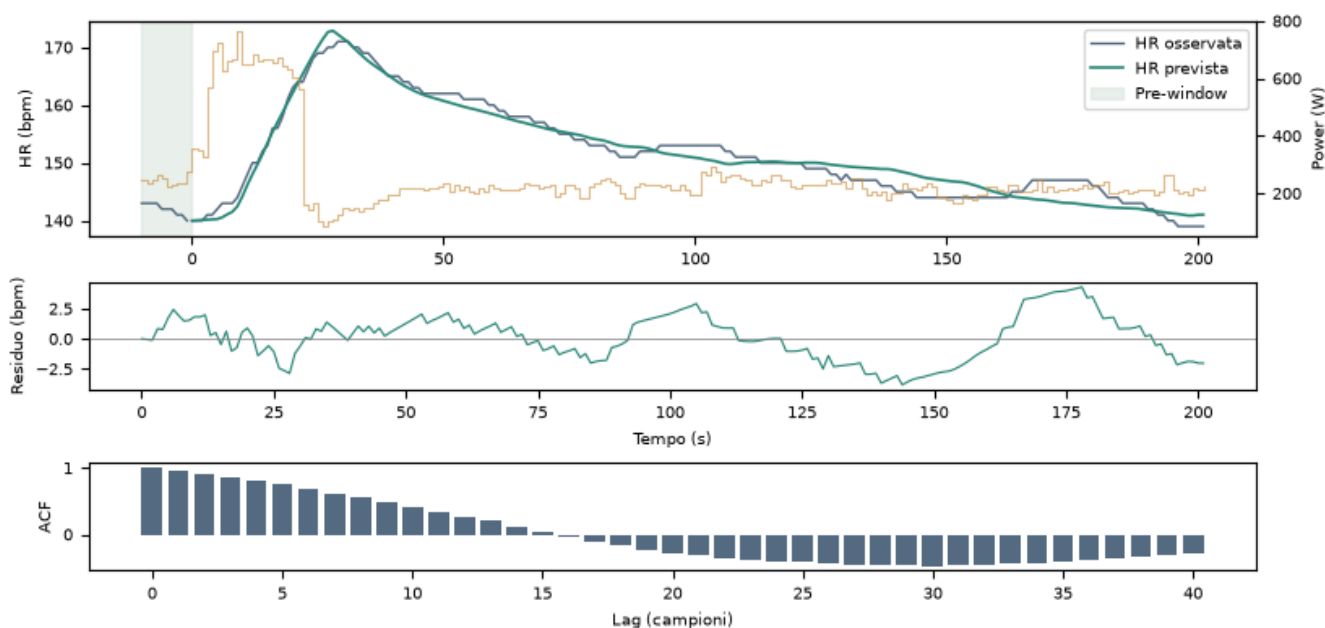
ID: 2432105a3e917cde7268373e | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=244.6 W; HR0=140 bpm.

HR(0) osservata fissata; B=146.03 bpm stimato; $x(0)=-6.0324$ bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.855	1.523	0.098652	0.9501	1.0268
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.098652	1.857	0.9547	0.75741	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.46717	0.039325	0.3901 / 0.54425	8.4176	no	no
L	5.4792	0.26729	4.9554 / 6.0031	4.8783	no	no
tau	108.8	10.133	88.938 / 128.66	9.3138	no	no
B	146.03	0.64774	144.76 / 147.3	0.44356	no	no

rho=1.7971; Jacobian rank=4/4; cond=2031.6; accordo multistart=no.

Correlazioni: {"K_L": -0.43096072250088036, "K_tau": 0.9900597367534187, "K_B": -0.9362266109979166, "L_tau": -0.4478916825758321, "L_B": 0.3378272838193231, "tau_B": -0.9220621344129979}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 695.1 / 13128

Griglia: 42441 coppie L/tau; SSE multistart originale: 695.1; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250604173939__laps_009-010.csv

ID: 26a8c904c169cd2064ccba9e | Stato: fitted

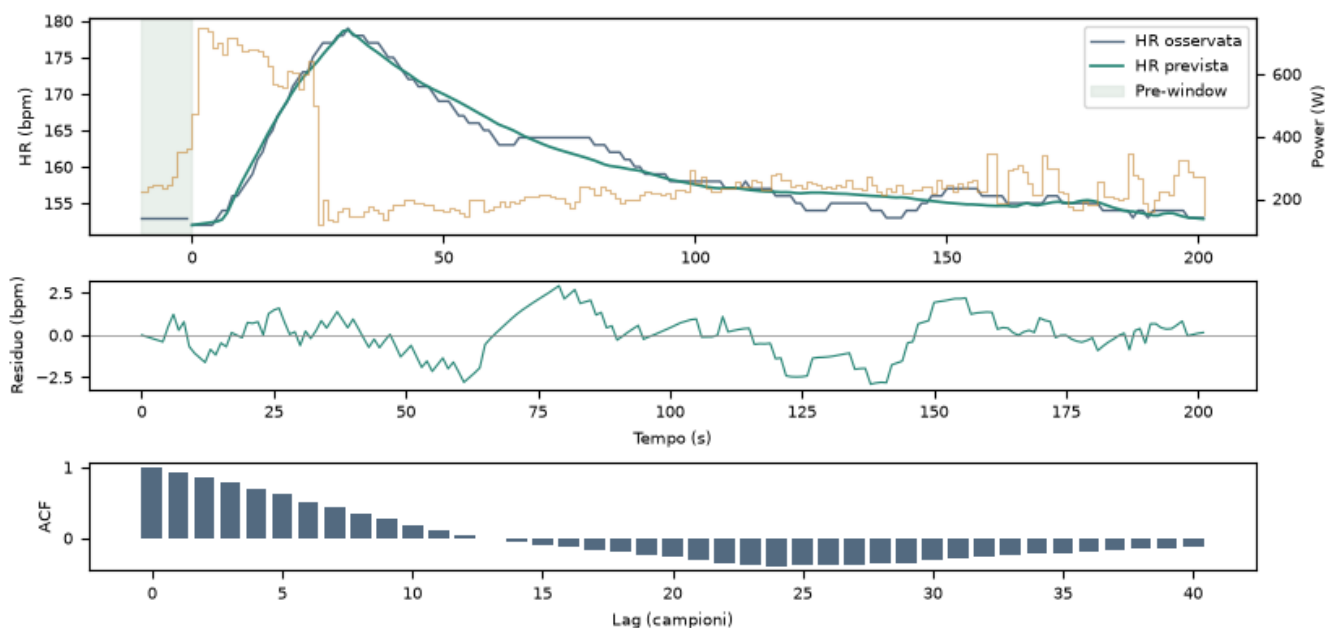
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=277.1 W; HR0=152 bpm.

HR(0) osservata fissata; B=162.3 bpm stimato; x(0)=-10.3 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.2476	0.96339	-0.0054132	0.97086	0.99135

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.0054132	1.2507	0.91783	0.61508



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.30691	0.016605	0.27436 / 0.33945	5.4104	no	no
L	5.6906	0.24241	5.2155 / 6.1657	4.2599	no	no
tau	101.55	6.2014	89.395 / 113.7	6.1068	no	no
B	162.3	0.15601	161.99 / 162.61	0.096127	no	no

rho=1.9233; Jacobian rank=4/4; cond=2124.1; accordo multistart=no.

Correlazioni: {"K_L": -0.484623942060502, "K_tau": 0.9843680537171852, "K_B": 0.30697283360760963, "L_tau": -0.49526412742423137, "L_B": -0.3418015048025866, "tau_B": 0.30409209953489336}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 314.41 / 10402

Griglia: 42441 coppie L/tau; SSE multistart originale: 314.41; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260321102409__laps_010-011.csv

ID: 3156c25d43f4239bdb596f18 | Stato: fitted

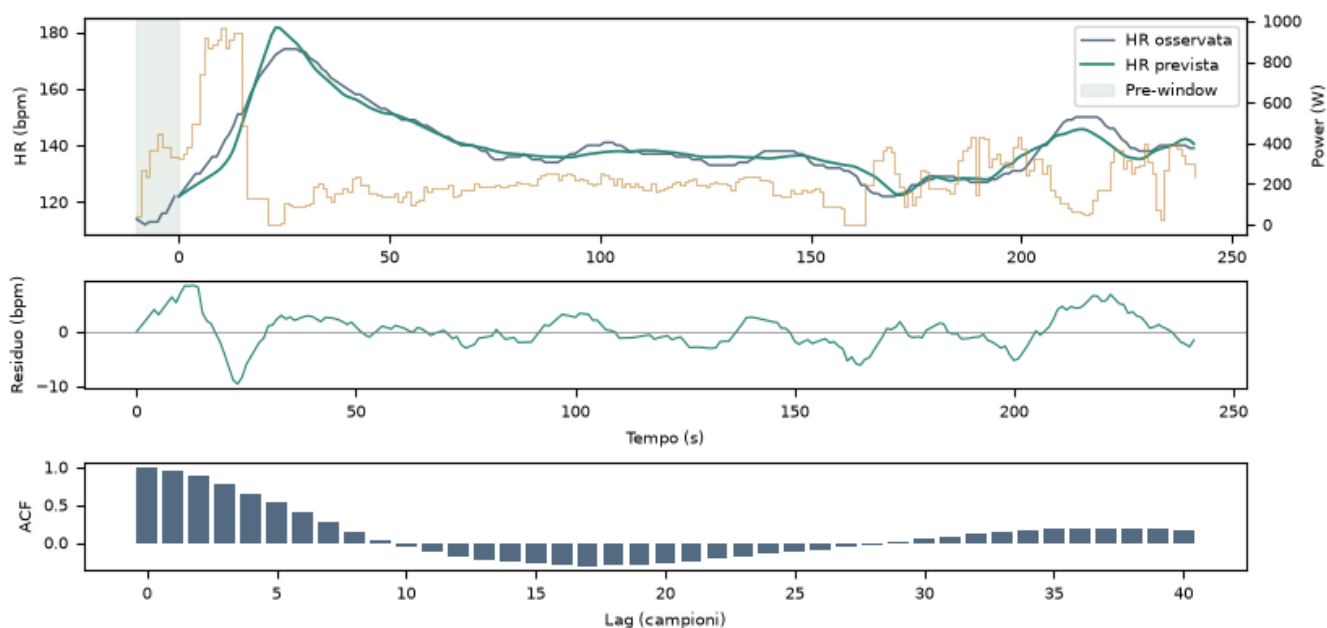
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=315.2 W; HR0=122 bpm.

HR(0) osservata fissata; B=174.64 bpm stimato; x(0)=-52.636 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.9947	2.2899	0.19281	0.93737	1.1464

residual_mean	residual_sd	ACF lag 1	ACF lag 5
0.19281	2.9947	0.95588	0.52961



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.36258	0.015235	0.33272 / 0.39244	4.2017	no	no
L	7.5017	0.19886	7.1119 / 7.8914	2.6509	no	no
tau	50.913	2.4533	46.105 / 55.721	4.8186	no	no
B	174.64	1.4893	171.72 / 177.55	0.8528	no	no

rho=4.5862; Jacobian rank=4/4; cond=1276.1; accordo multistart=no.

Correlazioni: {"K_L": -0.4917247202368018, "K_tau": 0.9502953906562079, "K_B": 0.9872344502534929, "L_tau": -0.48162420297888175, "L_B": -0.5056489514434682, "tau_B": 0.9410473098403985}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 2170.3 / 32647

Griglia: 42441 coppie L/tau; SSE multistart originale: 2170.3; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250709173010__laps_013-014.csv

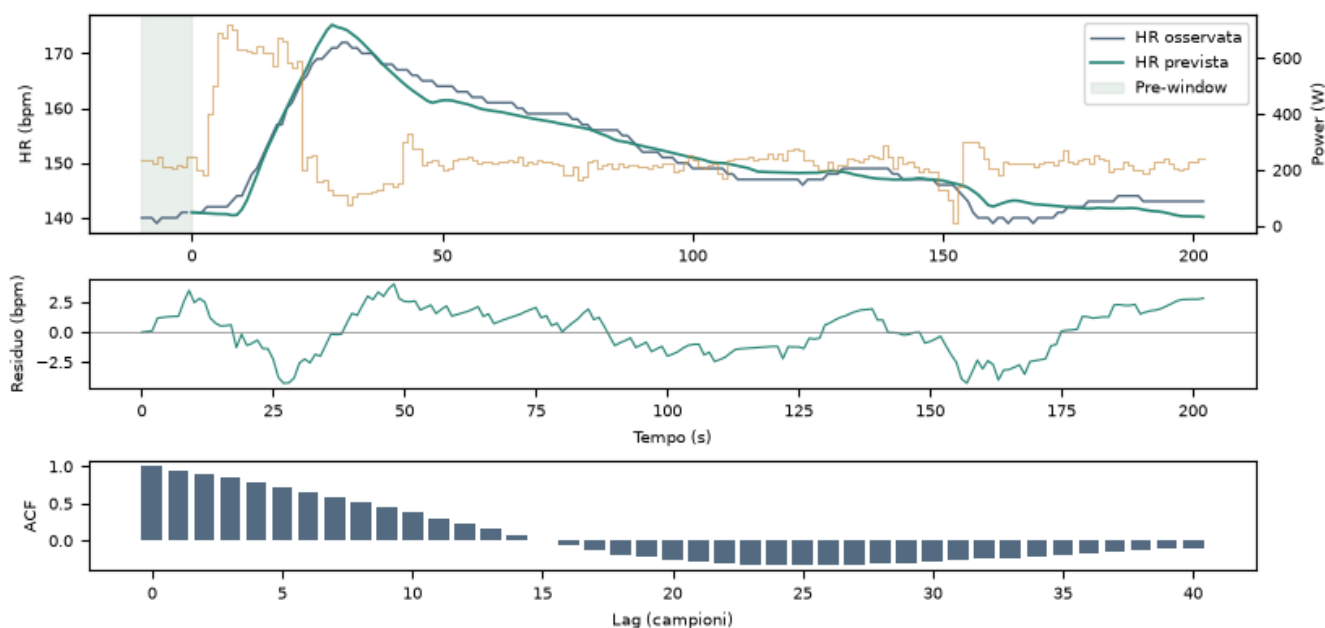
ID: 3650ea20551758dbe25b2e31 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=225.6 W; HR0=141 bpm.

HR(0) osservata fissata; B=134 bpm stimato; x(0)=6.9986 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.8831	1.6164	0.11215	0.95907	1.0622
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.11215	1.8844	0.95124	0.72025	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.61566	0.046679	0.52417 / 0.70715	7.582	no	no
L	5.7065	0.22298	5.2694 / 6.1435	3.9075	no	no
tau	117.47	9.8815	98.107 / 136.84	8.4116	no	no
B	134	1.5056	131.05 / 136.95	1.1236	no	no

rho=1.6709; Jacobian rank=4/4; cond=2000.5; accordo multistart=no.

Correlazioni: {"K_L": -0.3795354860543327, "K_tau": 0.9887967037197359, "K_B": -0.9877076644082157, "L_tau": -0.3994219925539753, "L_B": 0.3569296719746821, "tau_B": -0.9759560629931053}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 719.85 / 16868

Griglia: 42441 coppie L/tau; SSE multistart originale: 719.85; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250604173939__laps_003-004.csv

ID: 38bf5e52e0c89cac13997cfc | Stato: fitted

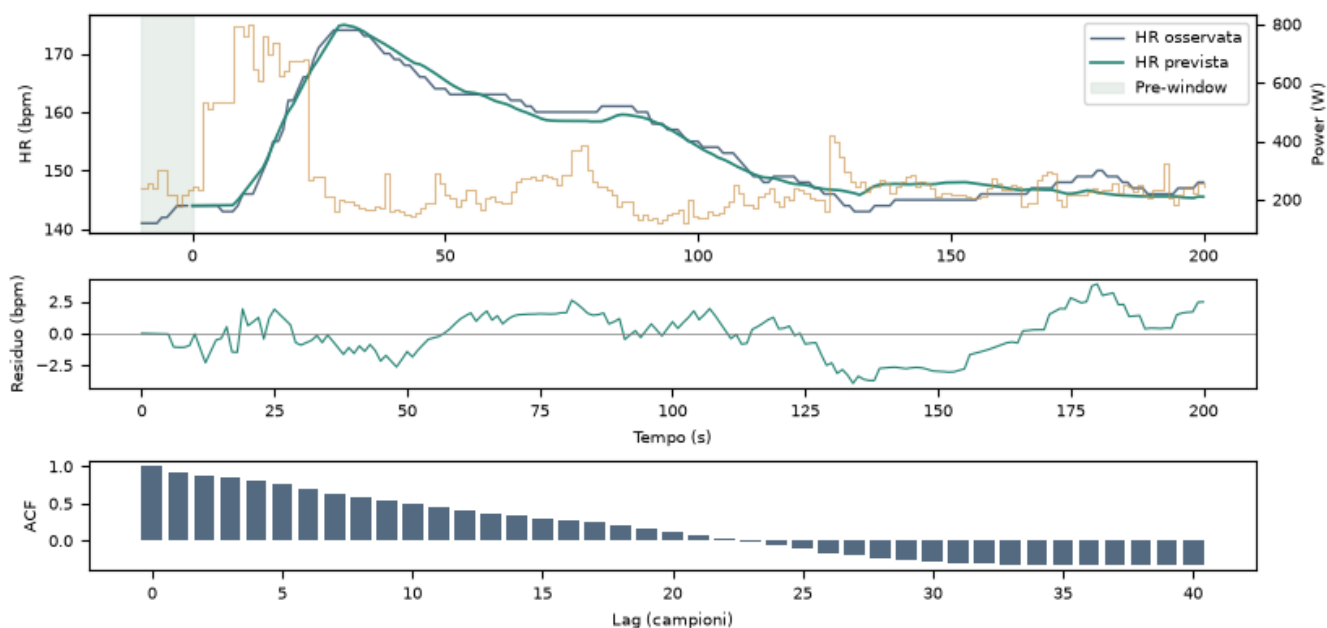
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=239.8 W; HR0=144 bpm.

HR(0) osservata fissata; B=144.98 bpm stimato; $x(0)=-0.98483$ bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.6873	1.3877	-0.04769	0.96551	0.99715

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.04769	1.6908	0.93202	0.7597



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.34552	0.0184	0.30946 / 0.38159	5.3253	no	no
L	6.251	0.25159	5.7579 / 6.7441	4.0248	no	no
tau	87.28	5.3418	76.81 / 97.75	6.1203	no	no
B	144.98	0.59418	143.82 / 146.15	0.40982	no	no

rho=2.2199; Jacobian rank=4/4; cond=1678.6; accordo multistart=no.

Correlazioni: {"K_L": -0.4694698273457047, "K_tau": 0.9787842300402829, "K_B": -0.9502995072305187, "L_tau": -0.4981912296268546, "L_B": 0.40750616587301586, "tau_B": -0.9303246771523583}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 572.22 / 15812

Griglia: 42441 coppie L/tau; SSE multistart originale: 572.22; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260501094312__laps_006-007.csv

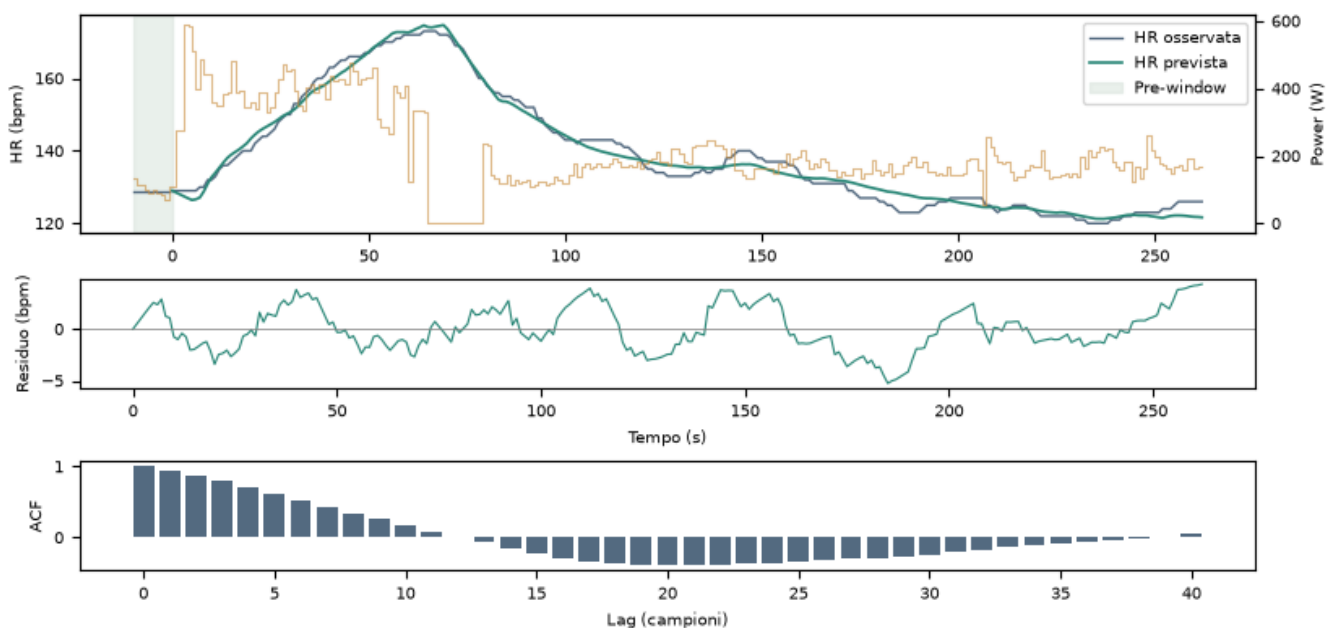
ID: 3f32a678f8f61184bbaa1dc0 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=100.1 W; HR0=129 bpm.

HR(0) osservata fissata; B=84.573 bpm stimato; x(0)=44.427 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.1015	1.7529	0.02657	0.98222	1.0084
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.02657	2.1053	0.93878	0.62055	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.4617	0.010929	0.44028 / 0.48312	2.3671	no	no
L	4.3011	0.2261	3.858 / 4.7443	5.2568	no	no
tau	83.551	2.612	78.431 / 88.67	3.1262	no	no
B	84.573	1.4019	81.825 / 87.321	1.6577	no	no

rho=3.0843; Jacobian rank=4/4; cond=2101.4; accordo multistart=no.

Correlazioni: {"K_L": -0.4141937304147623, "K_tau": 0.9450011643537646, "K_B": -0.9915598407834862, "L_tau": -0.51395810887052, "L_B": 0.4199927052659565, "tau_B": -0.9473165334545126}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1161.5 / 64546

Griglia: 42441 coppie L/tau; SSE multistart originale: 1161.5; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250604173939__laps_013-014.csv

ID: 4300d93cc4bc4e3f7e8a0770 | Stato: fitted

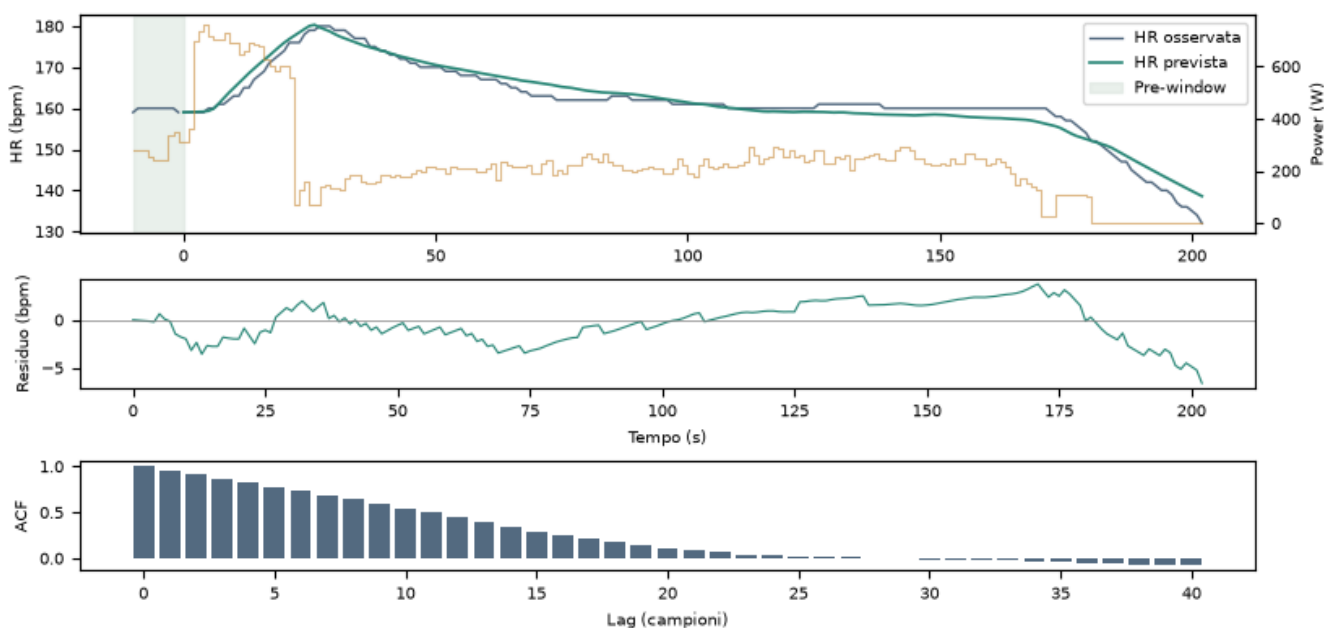
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=281.7 W; HR0=159 bpm.

HR(0) osservata fissata; B=166.21 bpm stimato; x(0)=-7.2136 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.0446	1.6858	-0.24098	0.94907	0.86919

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.24098	2.0353	0.94606	0.78228



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.47628	0.064525	0.34981 / 0.60275	13.548	no	no
L	3.6879	0.38127	2.9406 / 4.4352	10.338	no	no
tau	170.87	26.248	119.43 / 222.32	15.361	no	no
B	166.21	0.38492	165.46 / 166.97	0.23158	no	no

rho=1.1606; Jacobian rank=4/4; cond=3778.8; accordo multistart=no.

Correlazioni: {"K_L": -0.15649957691161173, "K_tau": 0.9928839625954322, "K_B": 0.2956902971946243, "L_tau": -0.18383421147061257, "L_B": -0.4216296579970326, "tau_B": 0.2691333130369706}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 848.59 / 16632

Griglia: 42441 coppie L/tau; SSE multistart originale: 848.59; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250604173939__laps_007-008.csv

ID: 435427565eac77eef857fb21 | Stato: fitted

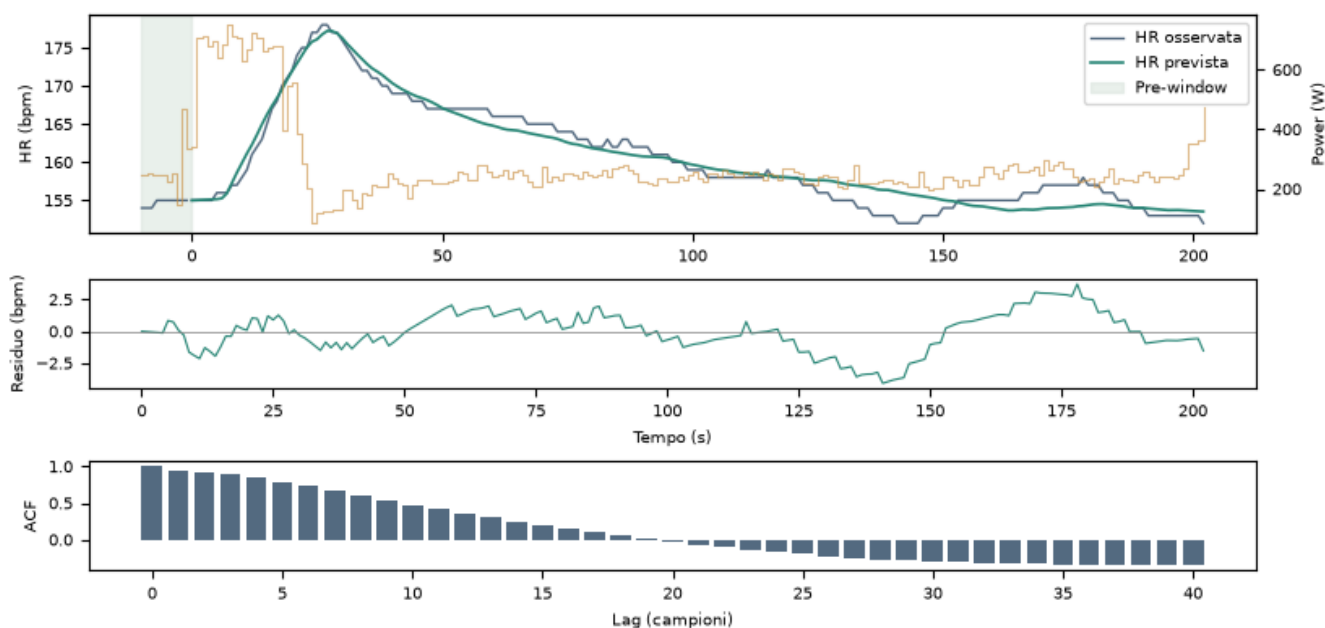
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=267.4 W; HR0=155 bpm.

HR(0) osservata fissata; B=157.42 bpm stimato; x(0)=-2.4219 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.5667	1.2413	-0.018496	0.94458	0.90961

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.018496	1.5705	0.94997	0.78378



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.23091	0.012543	0.20632 / 0.25549	5.4321	no	no
L	5.6581	0.31302	5.0446 / 6.2716	5.5322	no	no
tau	74.388	4.9469	64.693 / 84.084	6.6501	no	no
B	157.42	0.28698	156.86 / 157.98	0.1823	no	no

rho=2.6394; Jacobian rank=4/4; cond=1526.9; accordo multistart=no.

Correlazioni: {"K_L": -0.40517658084708996, "K_tau": 0.96394168027526, "K_B": -0.8240877322568634, "L_tau": -0.4322761611806583, "L_B": 0.253778370164689, "tau_B": -0.8176133368881334}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 498.29 / 8833.5

Griglia: 42441 coppie L/tau; SSE multistart originale: 498.29; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250715165154__laps_008-009.csv

ID: 4c896ae2c98d387828b17b0b | Stato: fitted

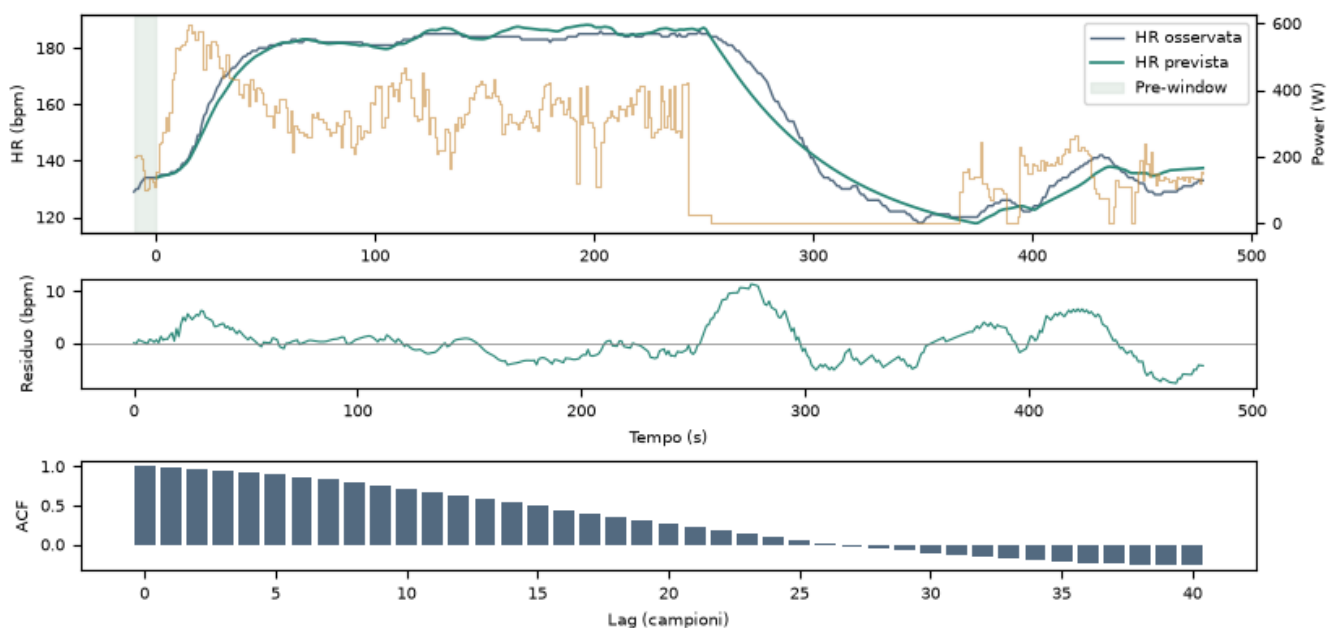
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=153.6 W; HR0=134 bpm.

HR(0) osservata fissata; B=145.29 bpm stimato; x(0)=-11.292 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
3.7877	2.8683	0.15386	0.97877	1.0354

residual_mean	residual_sd	ACF lag 1	ACF lag 5
0.15386	3.7885	0.98507	0.89787



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.23739	0.0024358	0.23262 / 0.24217	1.0261	no	no
L	7.48	0.46207	6.5744 / 8.3857	6.1775	no	no
tau	56.587	1.1423	54.348 / 58.826	2.0187	no	no
B	145.29	0.25241	144.8 / 145.79	0.17373	no	no

rho=8.3149; Jacobian rank=4/4; cond=866.92; accordo multistart=no.

Correlazioni: {"K_L": -0.4742960822895889, "K_tau": 0.7791908397233451, "K_B": -0.6394580689398369, "L_tau": -0.6545388823028621, "L_B": 0.23671306681528548, "tau_B": -0.5233771892593235}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 6871.9 / 3.172e+05

Griglia: 42441 coppie L/tau; SSE multistart originale: 6871.9; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260321102409__laps_006-007.csv

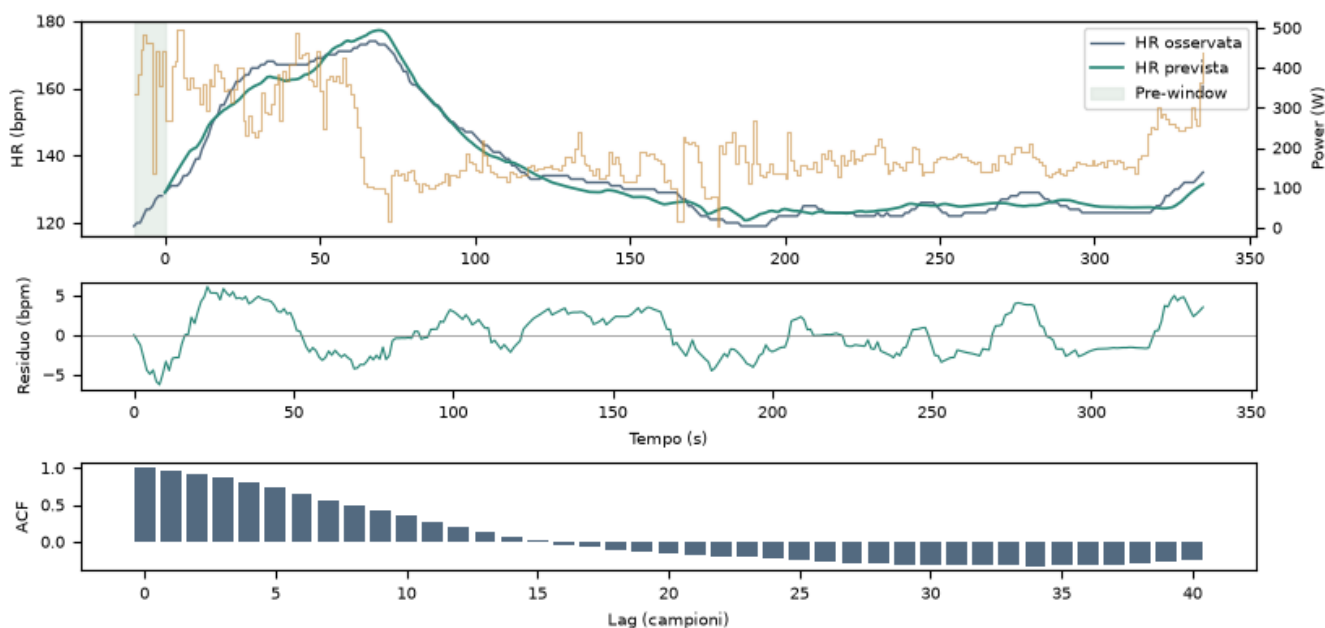
ID: 52ada156ddb27babecb7514d | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=395.1 W; HR0=129 bpm.

HR(0) osservata fissata; B=200 bpm stimato; x(0)=-71 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.6459	2.2907	0.087943	0.97608	1.0272
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.087943	2.6483	0.96559	0.72826	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.32403	0.0049479	0.31434 / 0.33373	1.527	no	no
L	8.264	0.36068	7.557 / 8.9709	4.3645	no	no
tau	42.199	0.92812	40.38 / 44.018	2.1994	no	no
B	200	1.0216	198 / 202	0.51081	si	si

rho=7.7428; Jacobian rank=4/4; cond=1728.6; accordo multistart=no.

Correlazioni: {"K_L": -0.6433493088625689, "K_tau": 0.8017550527508506, "K_B": 0.9855116296880355, "L_tau": -0.5350080480279399, "L_B": -0.675658034254848, "tau_B": 0.7738410353067712}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 2352.2 / 93347

Griglia: 42441 coppie L/tau; SSE multistart originale: 2352.2; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250709173010__laps_025-026.csv

ID: 52b3ceea9ad9f028edb15583 | Stato: fitted

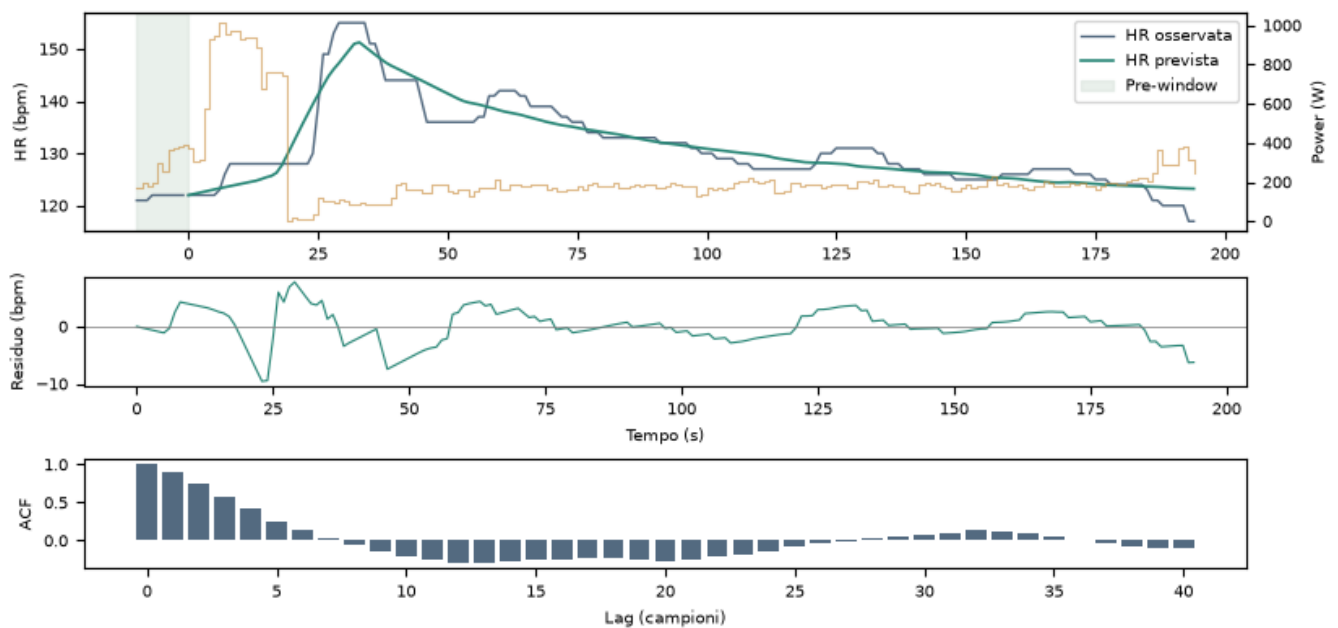
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=280.1 W; HR0=122 bpm.

HR(0) osservata fissata; B=144.63 bpm stimato; x(0)=-22.629 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.8923	2.1857	0.058119	0.86856	0.76971

residual_mean	residual_sd	ACF lag 1	ACF lag 5
0.058119	2.8992	0.89765	0.26013



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.26968	0.037551	0.19608 / 0.34328	13.924	no	no
L	13.535	0.47609	12.602 / 14.468	3.5175	no	no
tau	100.38	15.235	70.525 / 130.24	15.176	no	no
B	144.63	1.7191	141.26 / 148	1.1886	no	no

rho=1.7977; Jacobian rank=4/4; cond=3154.7; accordo multistart=no.

Correlazioni: {"K_L": -0.4704851636710838, "K_tau": 0.9875533591286975, "K_B": 0.9778943570389478, "L_tau": -0.47336475951893603, "L_B": -0.5112669700362246, "tau_B": 0.9679479481841045}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1631.3 / 11805

Griglia: 42441 coppie L/tau; SSE multistart originale: 1631.3; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260919065417__laps_004-005.csv

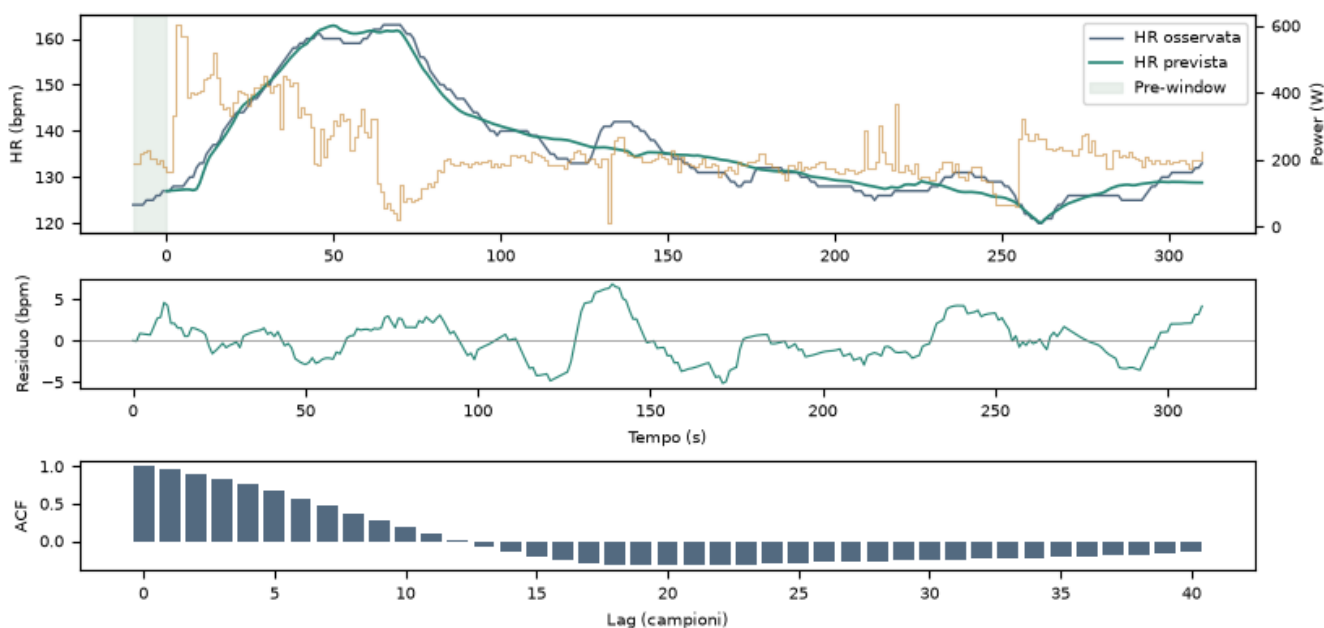
ID: 67b628ab6ed7b51e02b6a68f | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=203.2 W; HR0=127 bpm.

HR(0) osservata fissata; B=130.71 bpm stimato; x(0)=-3.7124 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.398	1.914	0.083822	0.9576	0.99284
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.083822	2.4004	0.96061	0.66754	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.34693	0.0082585	0.33074 / 0.36312	2.3804	no	no
L	6.5975	0.33553	5.9398 / 7.2551	5.0858	no	no
tau	62.655	2.0793	58.58 / 66.731	3.3186	no	no
B	130.71	0.23974	130.24 / 131.18	0.18341	no	no

rho=4.8424; Jacobian rank=4/4; cond=604.05; accordo multistart=no.

Correlazioni: {"K_L": -0.4383466765826009, "K_tau": 0.8964136529551033, "K_B": -0.7185267894203209, "L_tau": -0.5075922990676942, "L_B": 0.25851784895860025, "tau_B": -0.6855067423185321}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1788.4 / 41626

Griglia: 42441 coppie L/tau; SSE multistart originale: 1788.4; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250708171459__laps_004-005.csv

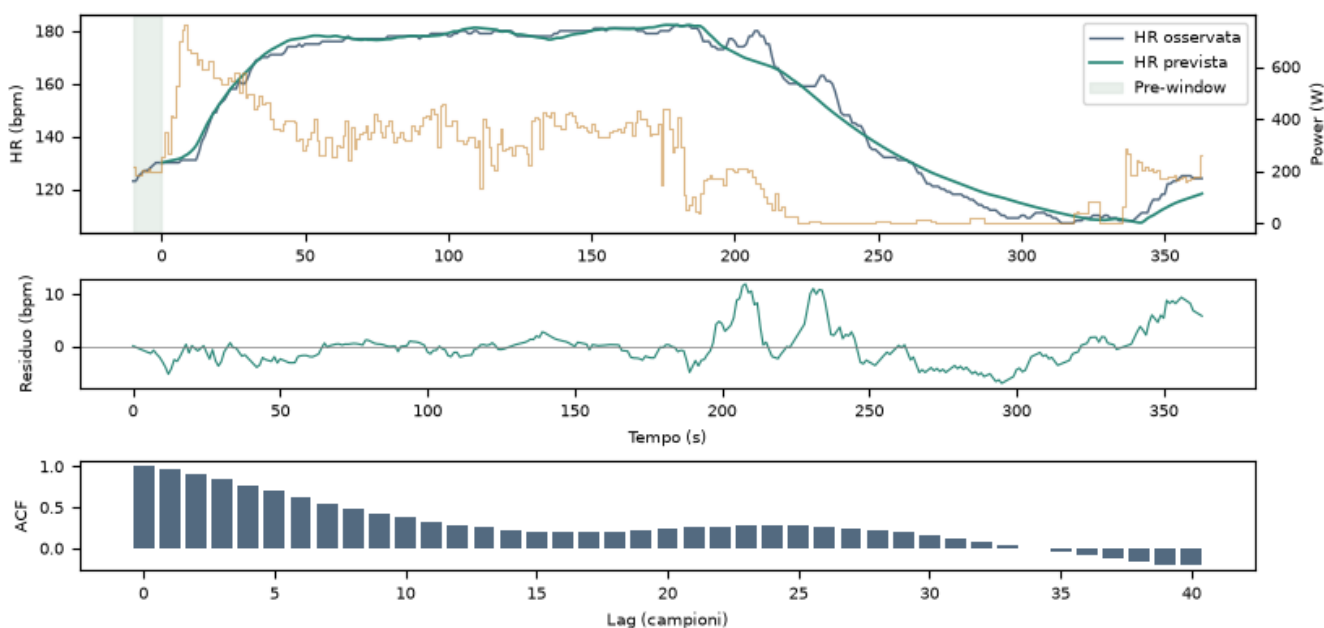
ID: 6f088b34503f7c77f7be1e0f | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=199.8 W; HR0=130 bpm.

HR(0) osservata fissata; B=145.09 bpm stimato; x(0)=-15.09 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
3.5507	2.5128	-0.19139	0.98394	1.0008
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.19139	3.5505	0.96161	0.6992	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.24573	0.0037239	0.23843 / 0.25303	1.5154	no	no
L	6.2919	0.50615	5.2998 / 7.2839	8.0446	no	no
tau	60.167	1.6356	56.961 / 63.373	2.7184	no	no
B	145.09	0.30868	144.48 / 145.69	0.21275	no	no

rho=5.9286; Jacobian rank=4/4; cond=1121.4; accordo multistart=no.

Correlazioni: {"K_L": -0.5953539354168322, "K_tau": 0.9067101017152399, "K_B": -0.6771121930166362, "L_tau": -0.7047063153602832, "L_B": 0.28355054641447147, "tau_B": -0.64749512176188}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 4589.2 / 2.8021e+05

Griglia: 42441 coppie L/tau; SSE multistart originale: 4589.2; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250709173010__laps_015-016.csv

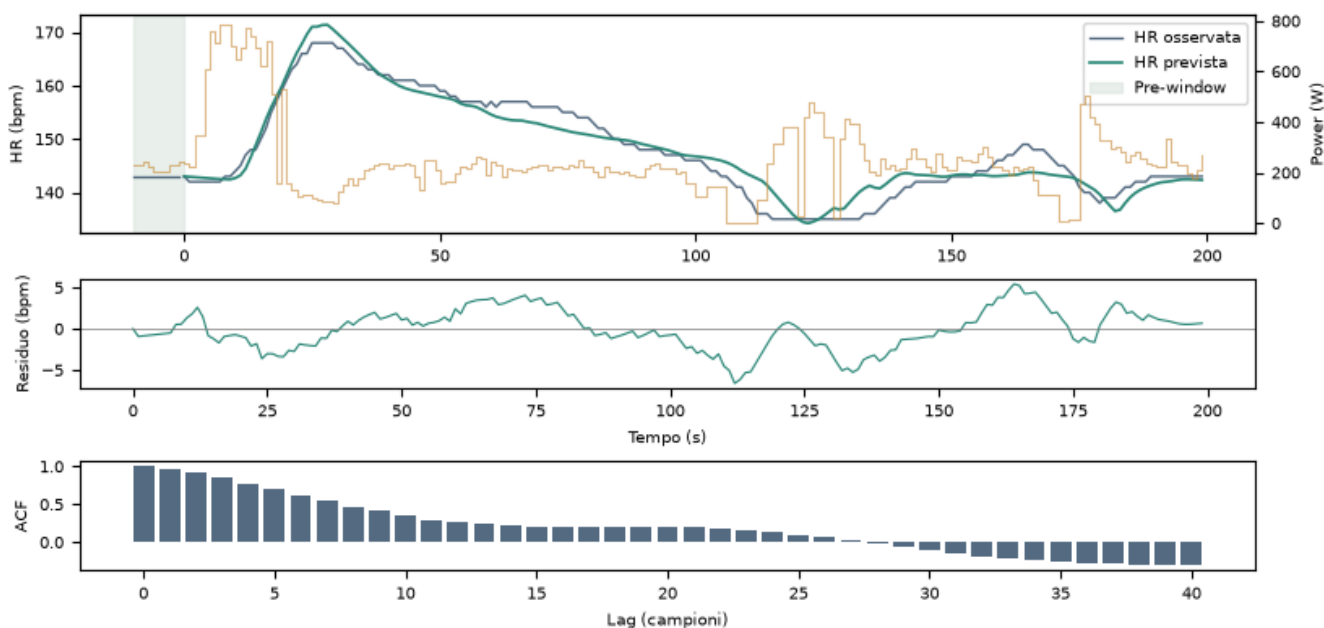
ID: 7b160af152a1c7415ebc1b83 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=224.5 W; HR0=143 bpm.

HR(0) osservata fissata; B=136.32 bpm stimato; x(0)=6.6765 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.4143	1.9423	-0.052527	0.9315	1.1262
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.052527	2.4198	0.96359	0.69512	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.43336	0.030086	0.37439 / 0.49233	6.9426	no	no
L	7.4238	0.25321	6.9275 / 7.9201	3.4108	no	no
tau	91.196	7.3183	76.852 / 105.54	8.0248	no	no
B	136.32	0.97678	134.41 / 138.24	0.71652	no	no

rho=2.1007; Jacobian rank=4/4; cond=1187.9; accordo multistart=no.

Correlazioni: {"K_L": -0.3421475315886167, "K_tau": 0.9679620264722613, "K_B": -0.9599692918411873, "L_tau": -0.36693204726969725, "L_B": 0.32084069051099173, "tau_B": -0.9423027126051187}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1165.8 / 16829

Griglia: 42441 coppie L/tau; SSE multistart originale: 1165.8; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260501094312__laps_008-010.csv

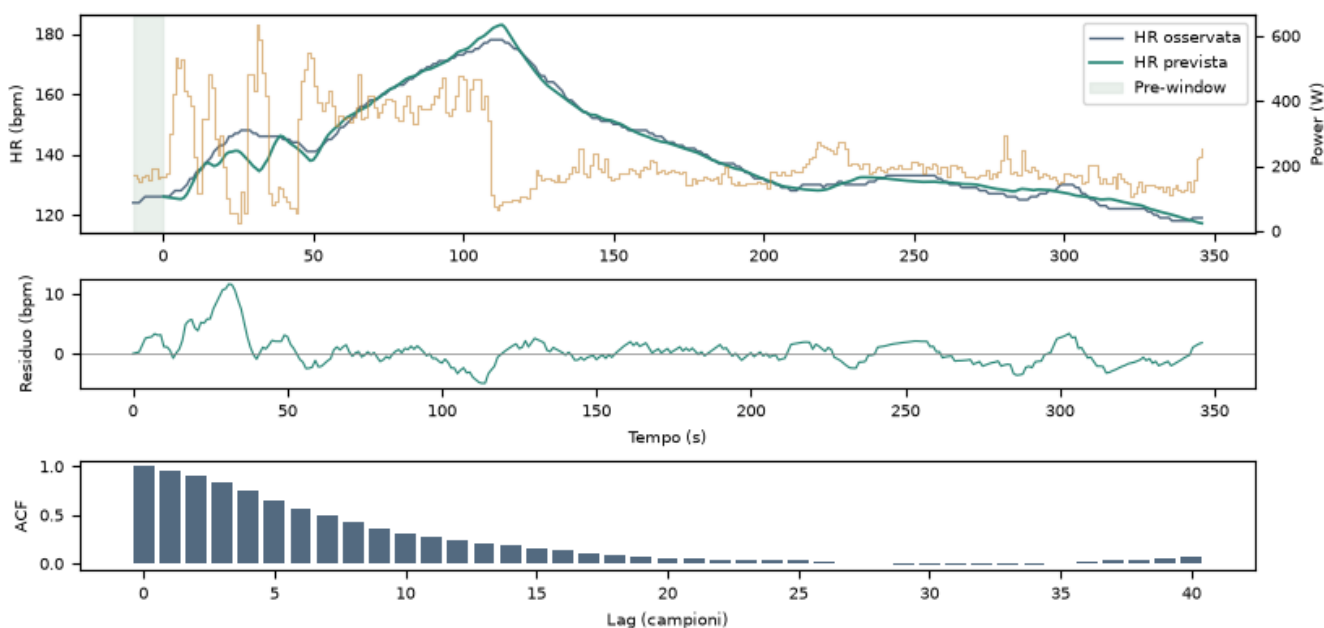
ID: 7da2631ee651e8807b7c0d7f | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=169.9 W; HR0=126 bpm.

HR(0) osservata fissata; B=116.91 bpm stimato; x(0)=9.0857 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.4577	1.6658	0.33499	0.97738	1.0962
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.33499	2.4382	0.96259	0.65714	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.39445	0.006024	0.38264 / 0.40626	1.5272	no	no
L	4.2808	0.22018	3.8493 / 4.7124	5.1434	no	no
tau	61.316	1.3381	58.693 / 63.938	2.1823	no	no
B	116.91	0.44419	116.04 / 117.78	0.37993	no	no

rho=5.5731; Jacobian rank=4/4; cond=742.14; accordo multistart=no.

Correlazioni: {"K_L": -0.3214797378140794, "K_tau": 0.8664918923813608, "K_B": -0.9359164344647449, "L_tau": -0.44688716738265594, "L_B": 0.2770605469974838, "tau_B": -0.8271367590842255}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 2095.9 / 90838

Griglia: 42441 coppie L/tau; SSE multistart originale: 2095.9; selezione: multistart.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250604173939__laps_017-018.csv

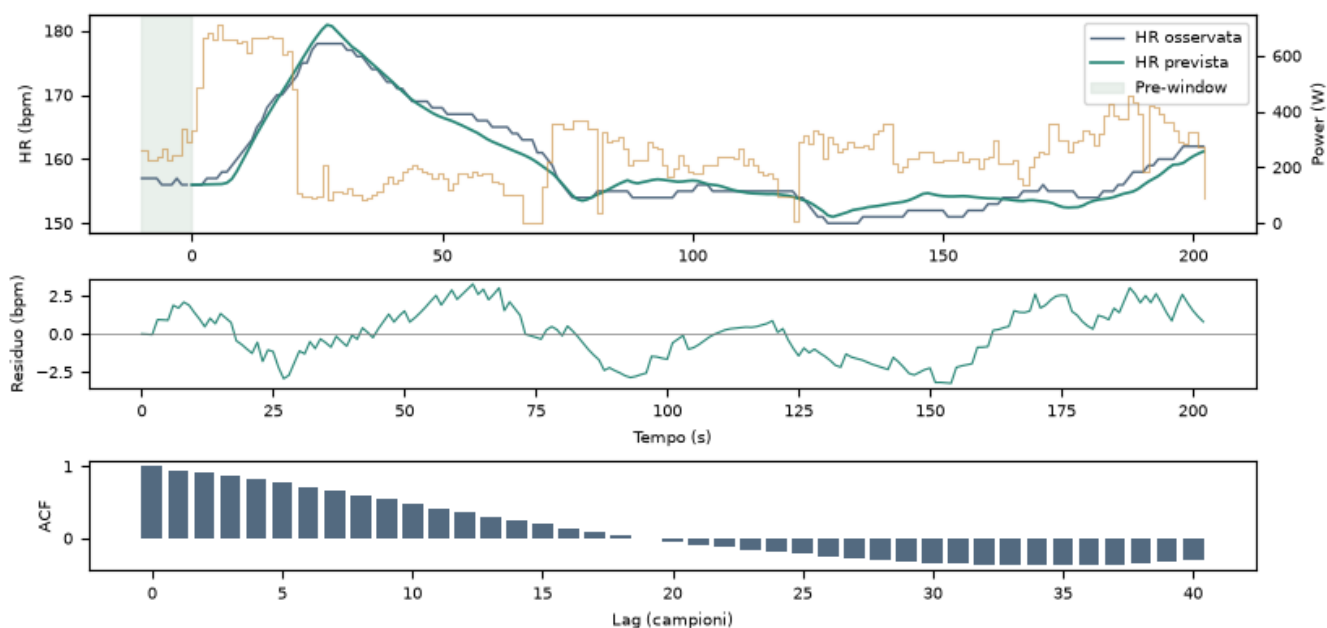
ID: 8019887eeae9da53fdc43a3c | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=256.9 W; HR0=156 bpm.

HR(0) osservata fissata; B=157.83 bpm stimato; $x(0)=-1.8344$ bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.6515	1.3946	0.036041	0.95431	1.0692
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.036041	1.6552	0.95282	0.7736	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.32623	0.017703	0.29153 / 0.36093	5.4265	no	no
L	6.4223	0.25122	5.9299 / 6.9147	3.9118	no	no
tau	91.417	5.6867	80.271 / 102.56	6.2206	no	no
B	157.83	0.22246	157.4 / 158.27	0.14094	no	no

rho=2.1394; Jacobian rank=4/4; cond=1211.2; accordo multistart=no.

Correlazioni: {"K_L": -0.4775068861740743, "K_tau": 0.9641033066044823, "K_B": -0.5517204436549158, "L_tau": -0.498209965142998, "L_B": 0.27342634475283506, "tau_B": -0.5795245118219894}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 553.68 / 13822

Griglia: 42441 coppie L/tau; SSE multistart originale: 553.68; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260321102409__laps_008-009.csv

ID: 89905a574f644b7823b2fd7e | Stato: fitted

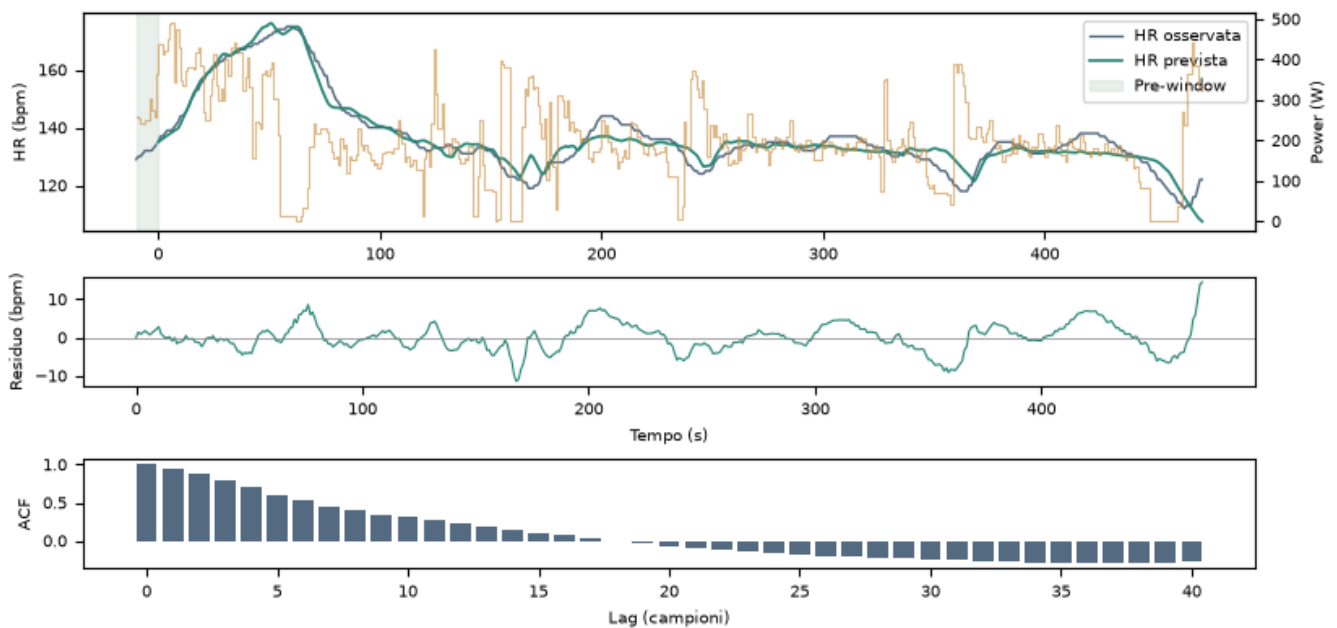
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=270.4 W; HR0=135 bpm.

HR(0) osservata fissata; B=160.73 bpm stimato; x(0)=-25.727 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
3.5939	2.7223	0.0034202	0.92731	1.0901

residual_mean	residual_sd	ACF lag 1	ACF lag 5
0.0034202	3.5977	0.95031	0.60851



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.35998	0.0071704	0.34592 / 0.37403	1.9919	no	no
L	9.3036	0.25227	8.8092 / 9.7981	2.7115	no	no
tau	43.756	1.3403	41.129 / 46.383	3.0631	no	no
B	160.73	0.50607	159.74 / 161.72	0.31486	no	no

rho=10.552; Jacobian rank=4/4; cond=626.45; accordo multistart=no.

Correlazioni: {"K_L": -0.2858655651190091, "K_tau": 0.7659846227396795, "K_B": 0.9308777357737256, "L_tau": -0.36978189850644405, "L_B": -0.31371574816875514, "tau_B": 0.6738782619564395}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 6096.3 / 83867

Griglia: 42441 coppie L/tau; SSE multistart originale: 6096.3; selezione: multistart.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250625105513__laps_023-024.csv

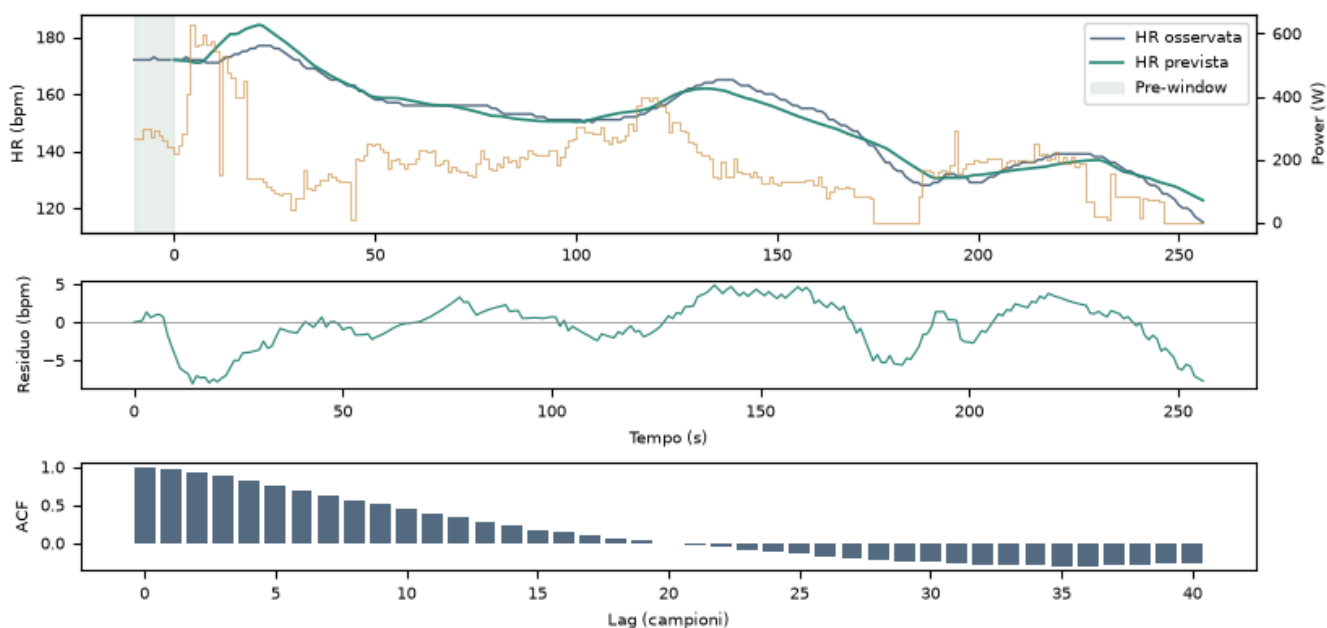
ID: 9d9868f8bc0dfed8b631f5cf | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=270.4 W; HR0=172 bpm.

HR(0) osservata fissata; B=165.42 bpm stimato; x(0)=6.585 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
3.0677	2.4062	-0.36025	0.95834	0.99396
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.36025	3.0524	0.96445	0.76052	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.29756	0.0073429	0.28317 / 0.31195	2.4677	no	no
L	3.423	0.37461	2.6888 / 4.1572	10.944	no	no
tau	55.806	2.3981	51.106 / 60.506	4.2972	no	no
B	165.42	0.48955	164.46 / 166.37	0.29595	no	no

rho=4.526; Jacobian rank=4/4; cond=878.06; accordo multistart=no.

Correlazioni: {"K_L": -0.13359493721595572, "K_tau": 0.6438039354737201, "K_B": 0.5762961811011085, "L_tau": -0.2500743870104717, "L_B": -0.10247545157904524, "tau_B": -0.11906723152617384}

Multistart: 16 tentativi, 15 convergenti; SSE min/max: 2418.6 / 15330

Griglia: 42441 coppie L/tau; SSE multistart originale: 2418.6; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250625105513__laps_013-016.csv

ID: 9da9a54d1702e65f30769703 | Stato: fitted

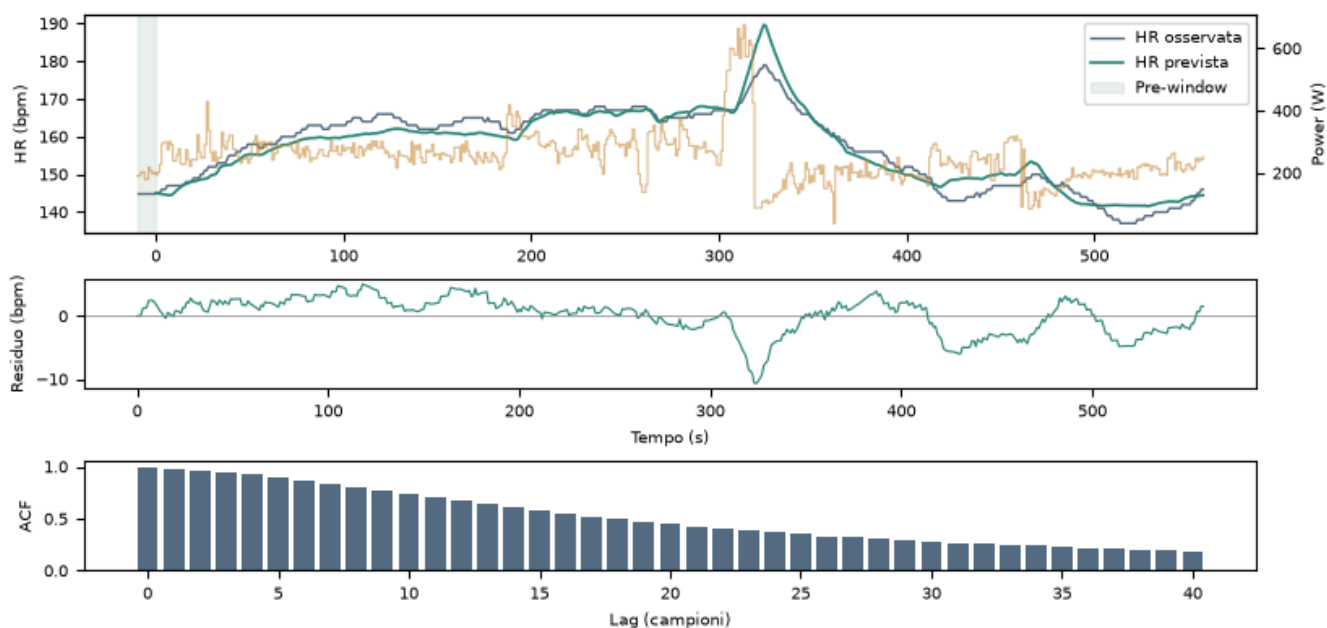
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=197 W; HR0=145 bpm.

HR(0) osservata fissata; B=141.05 bpm stimato; x(0)=3.9488 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.8496	2.2945	0.16513	0.92091	1.1447

residual_mean	residual_sd	ACF lag 1	ACF lag 5
0.16513	2.8473	0.98423	0.8963



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.29541	0.005512	0.28461 / 0.30621	1.8659	no	no
L	5.6583	0.30701	5.0566 / 6.2601	5.4258	no	no
tau	51.764	1.5734	48.68 / 54.848	3.0395	no	no
B	141.05	0.35506	140.36 / 141.75	0.25172	no	no

rho=10.67; Jacobian rank=4/4; cond=835.51; accordo multistart=no.

Correlazioni: {"K_L": -0.24581930496349577, "K_tau": 0.7419232872392403, "K_B": -0.9298002213870571, "L_tau": -0.36622016039285454, "L_B": 0.22492975461207937, "tau_B": -0.7041028245718226}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 4539.1 / 57237

Griglia: 42441 coppie L/tau; SSE multistart originale: 4539.1; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250709173010__laps_017-018.csv

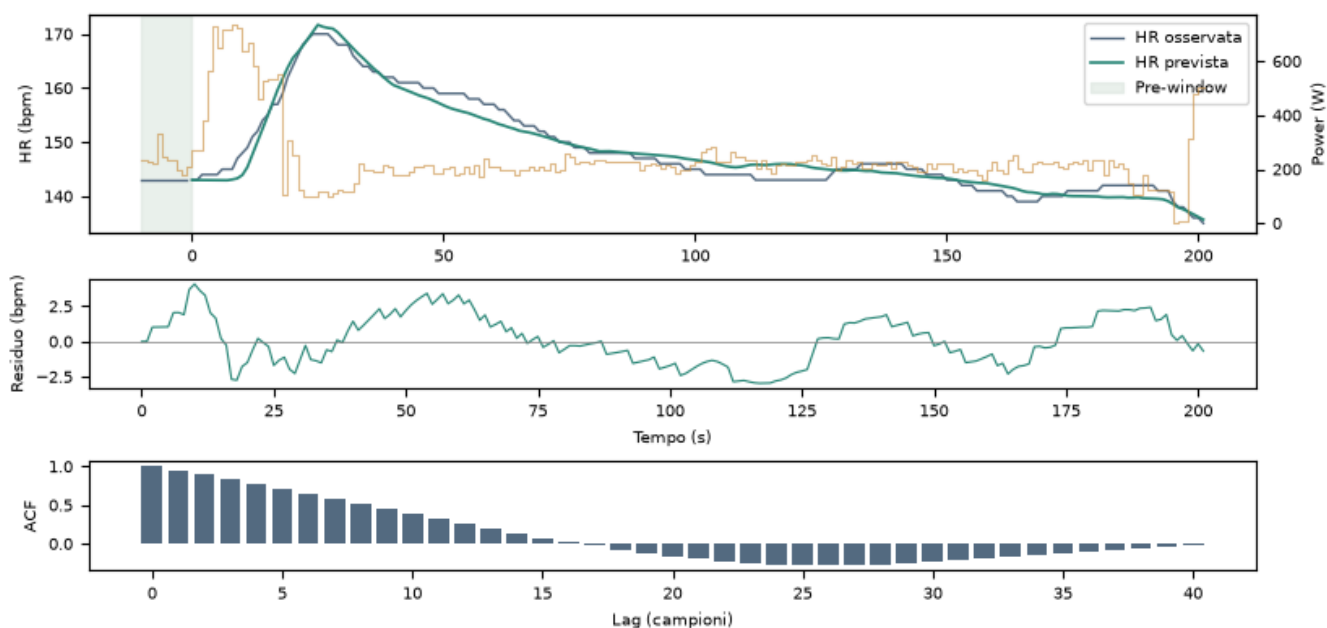
ID: 9e56c00128db1fd2a10dd426 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=231 W; HR0=143 bpm.

HR(0) osservata fissata; B=142.27 bpm stimato; x(0)=0.72683 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.7031	1.4248	0.12423	0.95996	1.029
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.12423	1.7028	0.94845	0.70709	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.46235	0.024698	0.41394 / 0.51076	5.3417	no	no
L	7.2693	0.21661	6.8447 / 7.6938	2.9798	no	no
tau	87.815	5.5767	76.885 / 98.745	6.3505	no	no
B	142.27	0.47496	141.34 / 143.2	0.33384	no	no

rho=2.2061; Jacobian rank=4/4; cond=1016.2; accordo multistart=no.

Correlazioni: {"K_L": -0.313862656464682, "K_tau": 0.9715281124957021, "K_B": -0.9155238763959959, "L_tau": -0.34549420609950454, "L_B": 0.23412764276420373, "tau_B": -0.9078437423952442}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 585.95 / 14505

Griglia: 42441 coppie L/tau; SSE multistart originale: 585.95; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260716153012__laps_004-005.csv

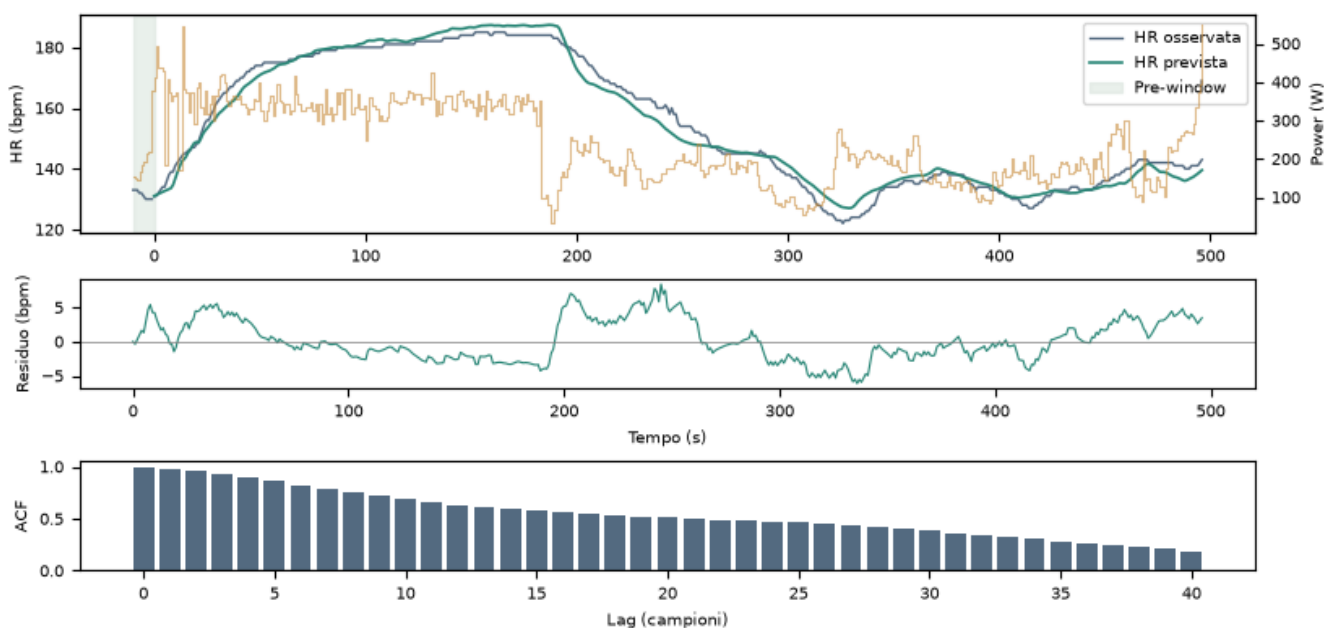
ID: a59ab2acbf2a3de3a7e865db | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=202.2 W; HR0=131 bpm.

HR(0) osservata fissata; B=146.04 bpm stimato; x(0)=-15.036 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
3.0078	2.4793	0.14999	0.97913	0.95885
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.14999	3.0071	0.97962	0.8585	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.29319	0.0024307	0.28843 / 0.29796	0.82904	no	no
L	8.4281	0.36282	7.7169 / 9.1392	4.3049	no	no
tau	42.006	0.77711	40.483 / 43.529	1.85	no	no
B	146.04	0.16665	145.71 / 146.36	0.11412	no	no

rho=11.607; Jacobian rank=4/4; cond=474.82; accordo multistart=no.

Correlazioni: {"K_L": -0.3427999456771636, "K_tau": 0.6486512766121837, "K_B": -0.4821964068324857, "L_tau": -0.5790153374862456, "L_B": 0.08565815868935638, "tau_B": -0.3152101735787702}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 4496.4 / 2.1085e+05

Griglia: 42441 coppie L/tau; SSE multistart originale: 4506.6; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260501094312__laps_012-013.csv

ID: a59f40db08815016b2c9a1b2 | Stato: fitted

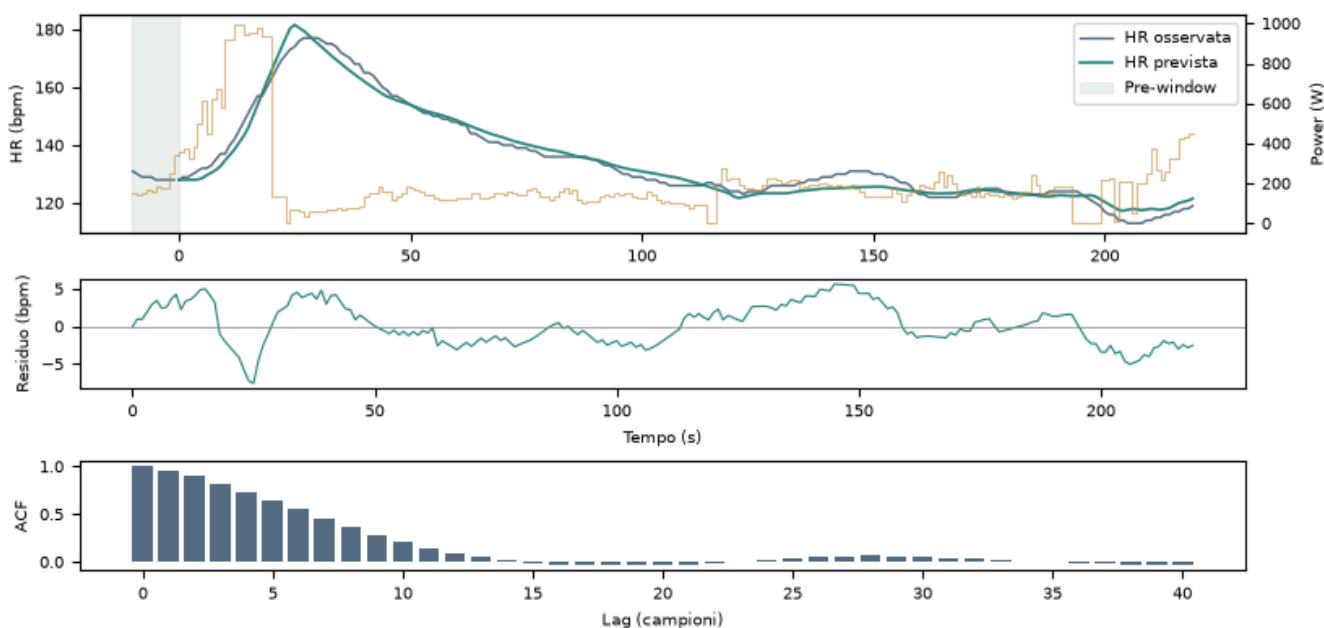
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=190.4 W; HR0=128 bpm.

HR(0) osservata fissata; B=129.58 bpm stimato; x(0)=-1.5751 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.6956	2.2468	0.15816	0.97275	1.0039

residual_mean	residual_sd	ACF lag 1	ACF lag 5
0.15816	2.6971	0.95607	0.64486



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.37699	0.010807	0.35581 / 0.39818	2.8666	no	no
L	4.5287	0.19524	4.1461 / 4.9114	4.3112	no	no
tau	66.774	2.4532	61.966 / 71.582	3.6738	no	no
B	129.58	0.34754	128.89 / 130.26	0.26822	no	no

rho=3.2119; Jacobian rank=4/4; cond=622.92; accordo multistart=no.

Correlazioni: {"K_L": -0.36962950149729734, "K_tau": 0.9299526222041814, "K_B": -0.6754565295750306, "L_tau": -0.40344479353780743, "L_B": 0.19602529984882663, "tau_B": -0.7032502299563512}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1598.6 / 58506

Griglia: 42441 coppie L/tau; SSE multistart originale: 1598.6; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250715165154__laps_004-005.csv

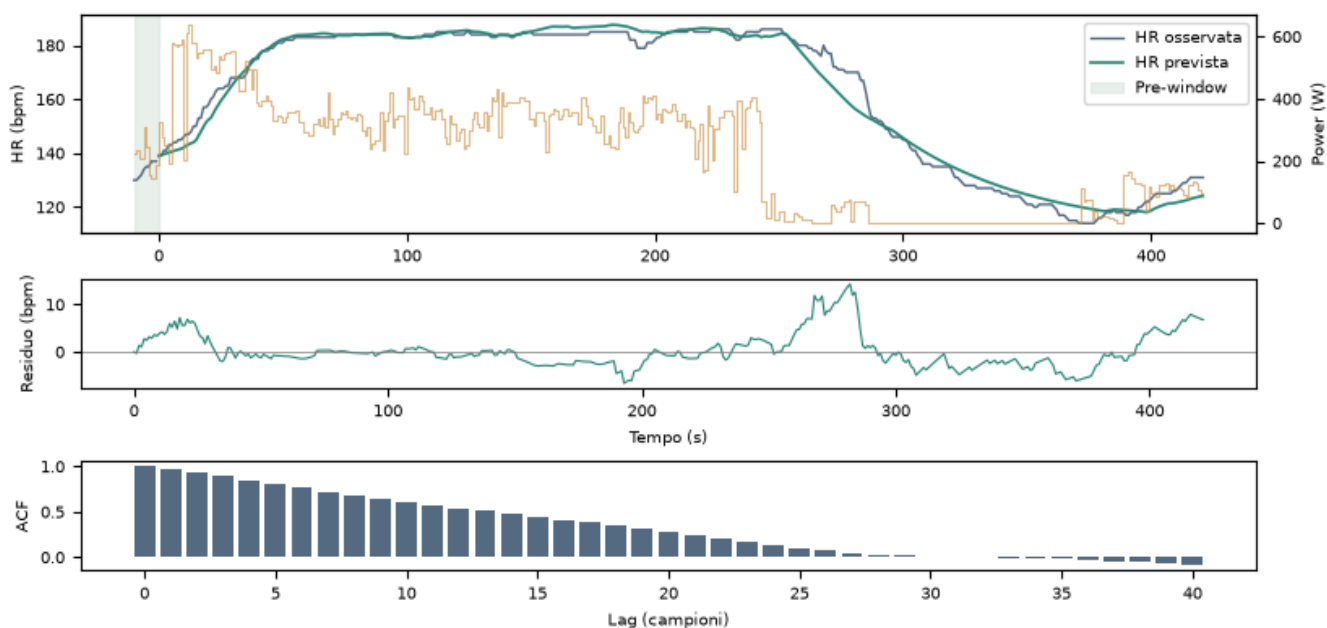
ID: a8e22f657d5002d3f9d9b849 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=206.9 W; HR0=139 bpm.

HR(0) osservata fissata; B=156.57 bpm stimato; $x(0)=-17.57$ bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
3.6529	2.5606	0.17013	0.98057	0.96551
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.17013	3.6532	0.97152	0.80862	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.22319	0.002885	0.21754 / 0.22885	1.2926	no	no
L	9.6124	0.54026	8.5535 / 10.671	5.6205	no	no
tau	55.105	1.4272	52.308 / 57.903	2.59	no	no
B	156.57	0.2503	156.08 / 157.06	0.15987	no	no

rho=7.4655; Jacobian rank=4/4; cond=1036.8; accordo multistart=no.

Correlazioni: {"K_L": -0.5020999284722072, "K_tau": 0.8573812438974536, "K_B": -0.5499682997733589, "L_tau": -0.6613684227813739, "L_B": 0.12796125969956906, "tau_B": -0.5065885036092386}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 5630.9 / 2.8508e+05

Griglia: 42441 coppie L/tau; SSE multistart originale: 5630.9; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250715165154__laps_002-003.csv

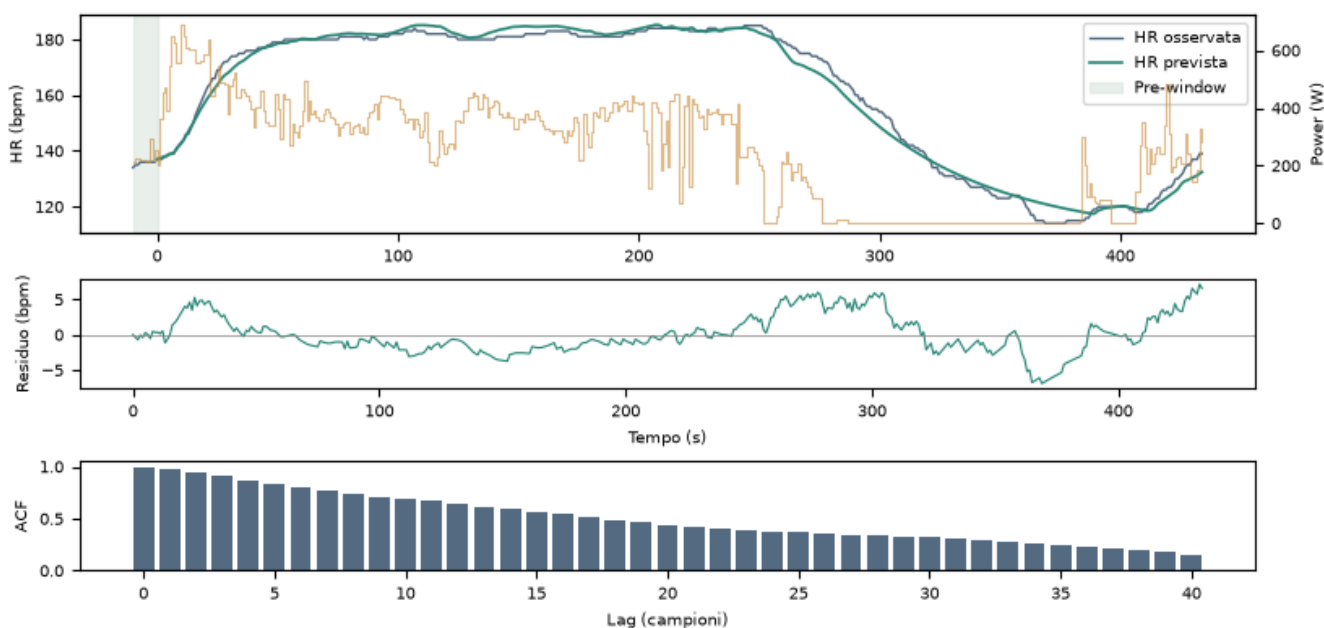
ID: aa444adc93a9b5093410ff88 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=229.3 W; HR0=137 bpm.

HR(0) osservata fissata; B=157.21 bpm stimato; x(0)=-20.211 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.7257	2.1128	0.072246	0.98848	0.95577
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.072246	2.7279	0.96927	0.83606	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.21028	0.0020708	0.20622 / 0.21434	0.98477	no	no
L	5.4194	0.40159	4.6323 / 6.2065	7.4102	no	no
tau	57.019	1.1015	54.86 / 59.178	1.9318	no	no
B	157.21	0.17366	156.87 / 157.55	0.11046	no	no

rho=7.5164; Jacobian rank=4/4; cond=1118.3; accordo multistart=no.

Correlazioni: {"K_L": -0.5494940608158349, "K_tau": 0.8651693569210163, "K_B": -0.482916655067838, "L_tau": -0.6783710118970372, "L_B": 0.14600490825682735, "tau_B": -0.46399569641186}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 3231.8 / 2.7537e+05

Griglia: 42441 coppie L/tau; SSE multistart originale: 3231.8; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250709173010__laps_027-028.csv

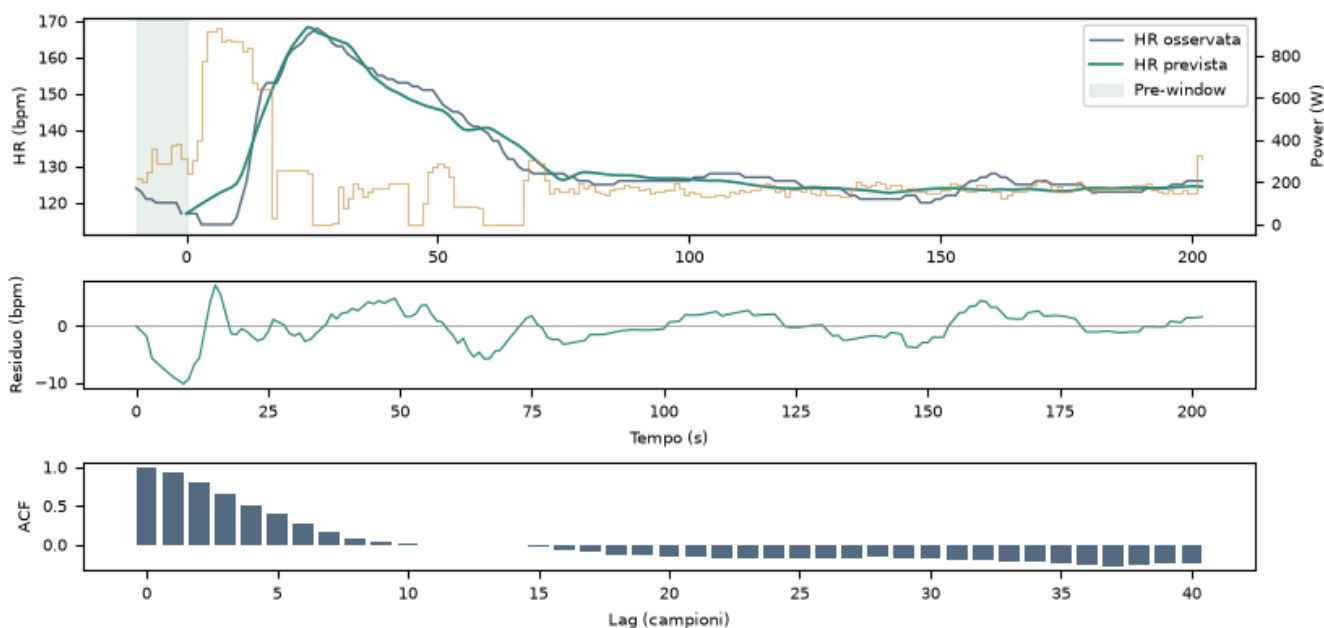
ID: b1240541ad46242fefe5003d | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=297.2 W; HR0=117 bpm.

HR(0) osservata fissata; B=153.91 bpm stimato; x(0)=-36.906 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.8672	2.1945	-0.28228	0.95476	0.95009
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.28228	2.8604	0.93413	0.38901	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.23378	0.0081513	0.21781 / 0.24976	3.4867	no	no
L	7.4658	0.26065	6.9549 / 7.9767	3.4912	no	no
tau	39.628	1.7317	36.234 / 43.022	4.3698	no	no
B	153.91	0.80679	152.32 / 155.49	0.52421	no	no

rho=4.909; Jacobian rank=4/4; cond=957.72; accordo multistart=no.

Correlazioni: {"K_L": -0.526313795755423, "K_tau": 0.9232715063683816, "K_B": 0.9518296315215952, "L_tau": -0.5311981660439675, "L_B": -0.5533233191108612, "tau_B": 0.8738485977320766}

Multistart: 16 tentativi, 15 convergenti; SSE min/max: 1668.9 / 35371

Griglia: 42441 coppie L/tau; SSE multistart originale: 1668.9; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250604173939__laps_011-012.csv

ID: b157a96f16e7f7750809b8dd | Stato: fitted

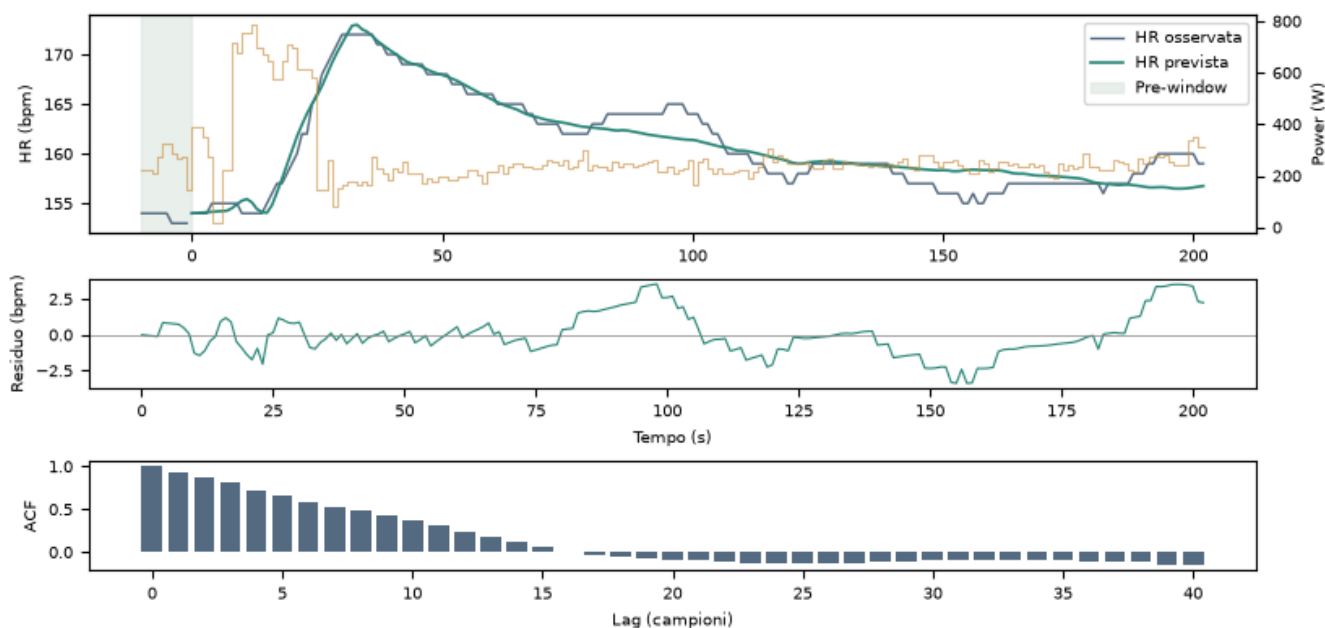
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=256.4 W; HR0=154 bpm.

HR(0) osservata fissata; B=158.32 bpm stimato; x(0)=-4.3204 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.4611	1.0789	-6.0604e-05	0.90831	1.0544

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-6.0604e-05	1.4647	0.93617	0.65562



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.32457	0.041865	0.24252 / 0.40662	12.898	no	no
L	7.4298	0.32342	6.7959 / 8.0637	4.353	no	no
tau	112.46	15.407	82.261 / 142.66	13.7	no	no
B	158.32	0.47174	157.4 / 159.24	0.29797	no	no

rho=1.7301; Jacobian rank=4/4; cond=3133.5; accordo multistart=no.

Correlazioni: {"K_L": -0.45629666357622, "K_tau": 0.9911725204061805, "K_B": -0.9223996381545184, "L_tau": -0.46896951921944435, "L_B": 0.36784263746143864, "tau_B": -0.9041552740917587}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 433.36 / 4263.4

Griglia: 42441 coppie L/tau; SSE multistart originale: 433.36; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260321102409__laps_004-005.csv

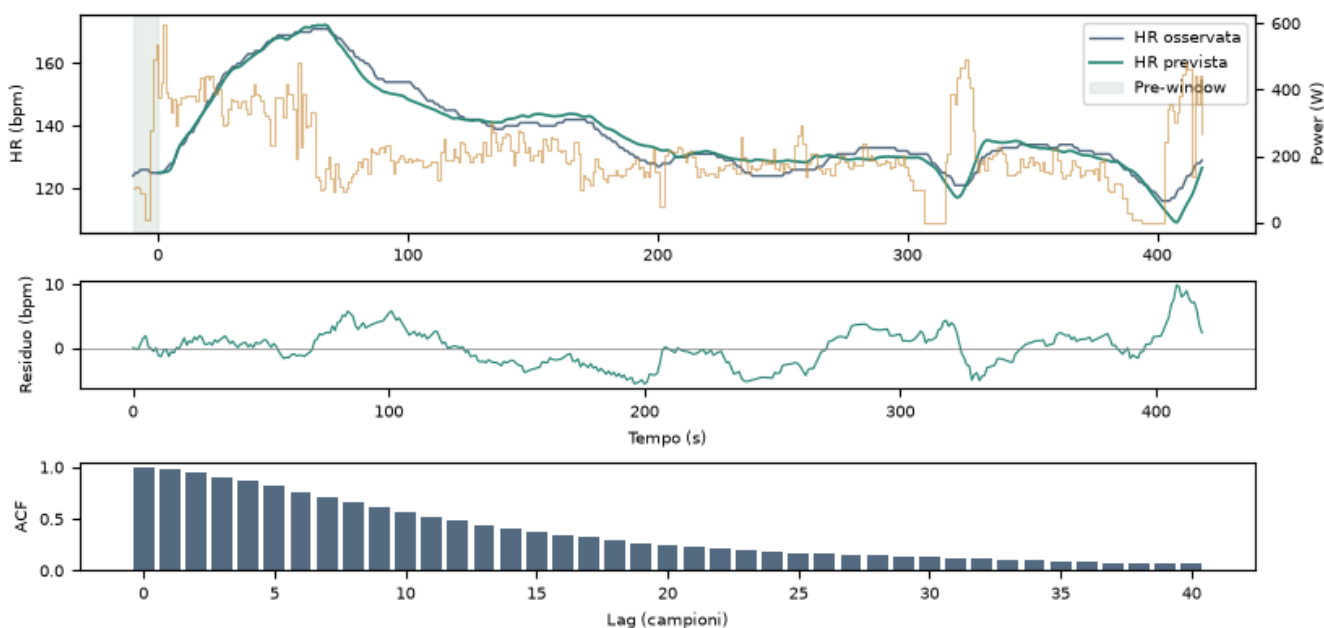
ID: b174858c39c7288152d48311 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=182.5 W; HR0=125 bpm.

HR(0) osservata fissata; B=129.97 bpm stimato; x(0)=-4.9748 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.8694	2.2928	0.065311	0.95792	1.149
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.065311	2.872	0.97646	0.81466	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.33854	0.0053353	0.32809 / 0.349	1.576	no	no
L	4.6038	0.22498	4.1628 / 5.0447	4.8868	no	no
tau	45.45	1.1473	43.201 / 47.699	2.5243	no	no
B	129.97	0.21361	129.56 / 130.39	0.16435	no	no

rho=9.0956; Jacobian rank=4/4; cond=402.68; accordo multistart=no.

Correlazioni: {"K_L": -0.3128835104222865, "K_tau": 0.8066245144118598, "K_B": -0.6828090591976214, "L_tau": -0.4120568872865266, "L_B": 0.180797696836591, "tau_B": -0.6085283493474268}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 3449.7 / 81084

Griglia: 42441 coppie L/tau; SSE multistart originale: 3449.7; selezione: multistart.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250715165154__laps_006-007.csv

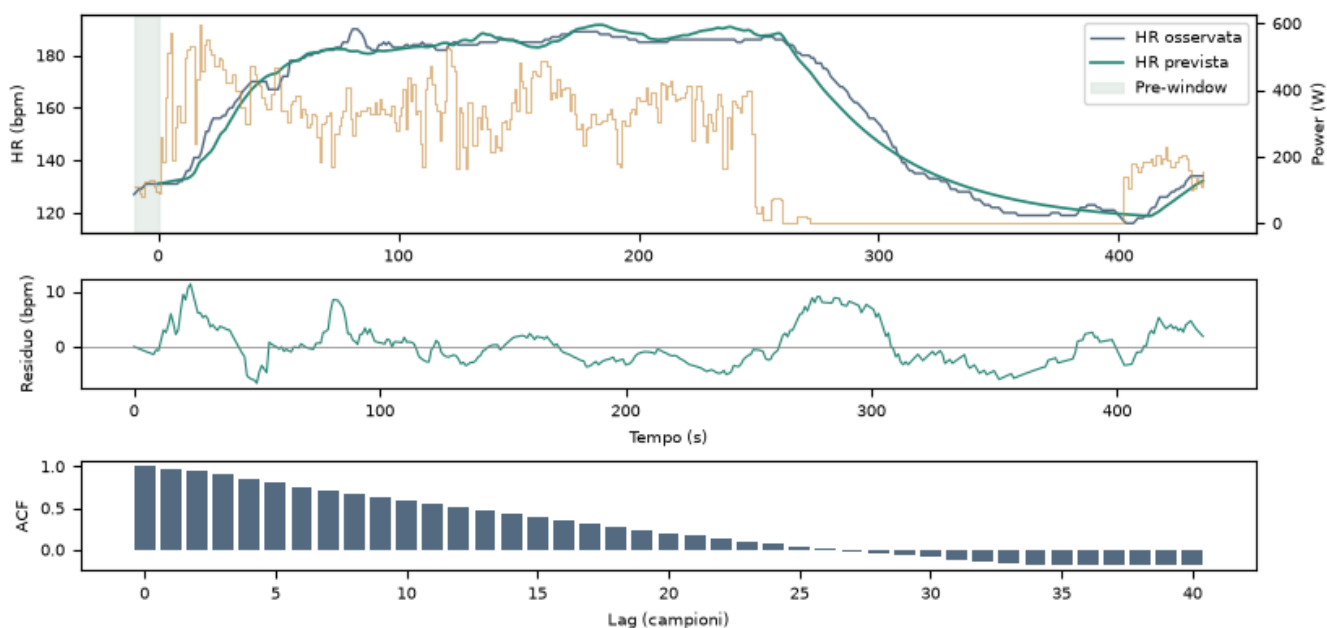
ID: b17c31f0a310602b50825ad1 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=111.3 W; HR0=131 bpm.

HR(0) osservata fissata; B=139.68 bpm stimato; x(0)=-8.6837 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
3.6782	2.9098	0.17225	0.98141	0.98468
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.17225	3.6784	0.97475	0.80576	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.21033	0.0022433	0.20594 / 0.21473	1.0666	no	no
L	11.442	0.45543	10.549 / 12.334	3.9804	no	no
tau	44.974	1.0545	42.907 / 47.041	2.3447	no	no
B	139.68	0.3395	139.02 / 140.35	0.24305	no	no

rho=9.4179; Jacobian rank=4/4; cond=1064.6; accordo multistart=no.

Correlazioni: {"K_L": -0.505961578332629, "K_tau": 0.7894495216701777, "K_B": -0.8192087703408627, "L_tau": -0.7031962358480441, "L_B": 0.380182232933752, "tau_B": -0.6697326736865524}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 5898.8 / 3.0469e+05

Griglia: 42441 coppie L/tau; SSE multistart originale: 5910.9; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260501094312__laps_004-005.csv

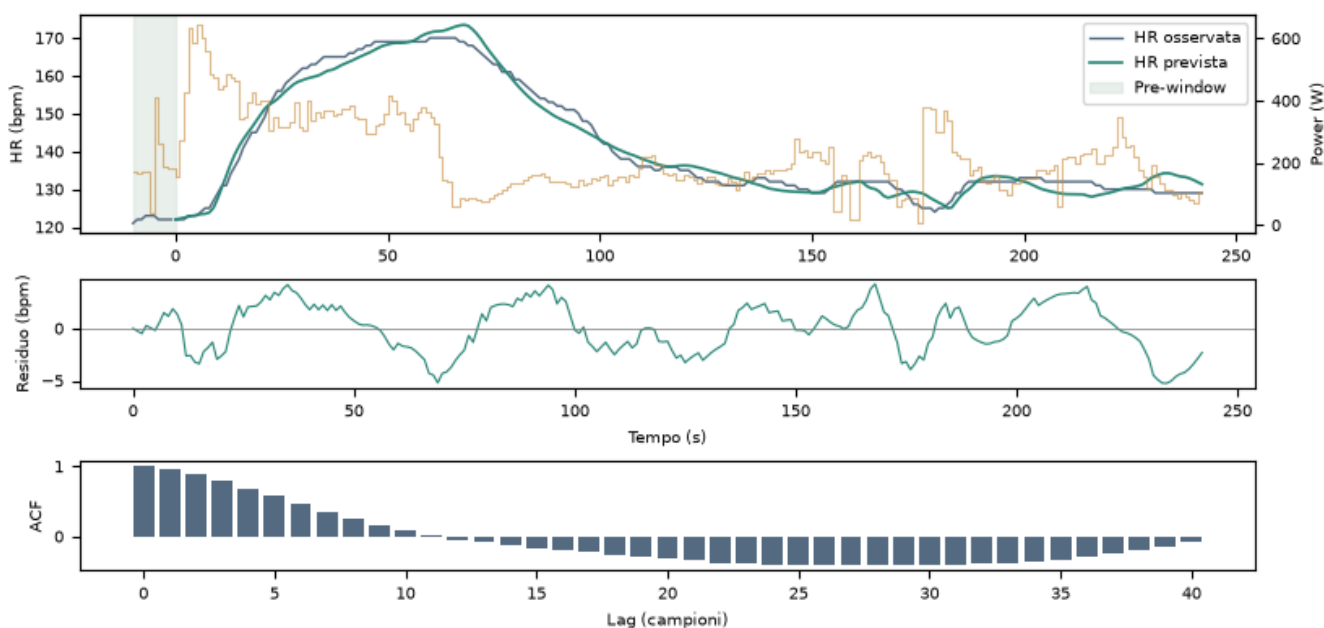
ID: b1b4978930221b9f8bb43aba | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=194.8 W; HR0=122 bpm.

HR(0) osservata fissata; B=133.9 bpm stimato; x(0)=-11.898 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.3238	1.9604	0.043716	0.97664	1.0722
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.043716	2.3282	0.95276	0.57554	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.33316	0.0067981	0.31983 / 0.34648	2.0405	no	no
L	6.5099	0.24969	6.0205 / 6.9993	3.8355	no	no
tau	47.567	1.2988	45.021 / 50.112	2.7305	no	no
B	133.9	0.25228	133.4 / 134.39	0.18841	no	no

rho=4.9507; Jacobian rank=4/4; cond=499.96; accordo multistart=no.

Correlazioni: {"K_L": -0.4994278509332096, "K_tau": 0.9078745180132344, "K_B": -0.6974666155475733, "L_tau": -0.5734842647507302, "L_B": 0.26068813519803713, "tau_B": -0.6313946909534728}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1312.2 / 53363

Griglia: 42441 coppie L/tau; SSE multistart originale: 1333.8; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260716153012__laps_008-009.csv

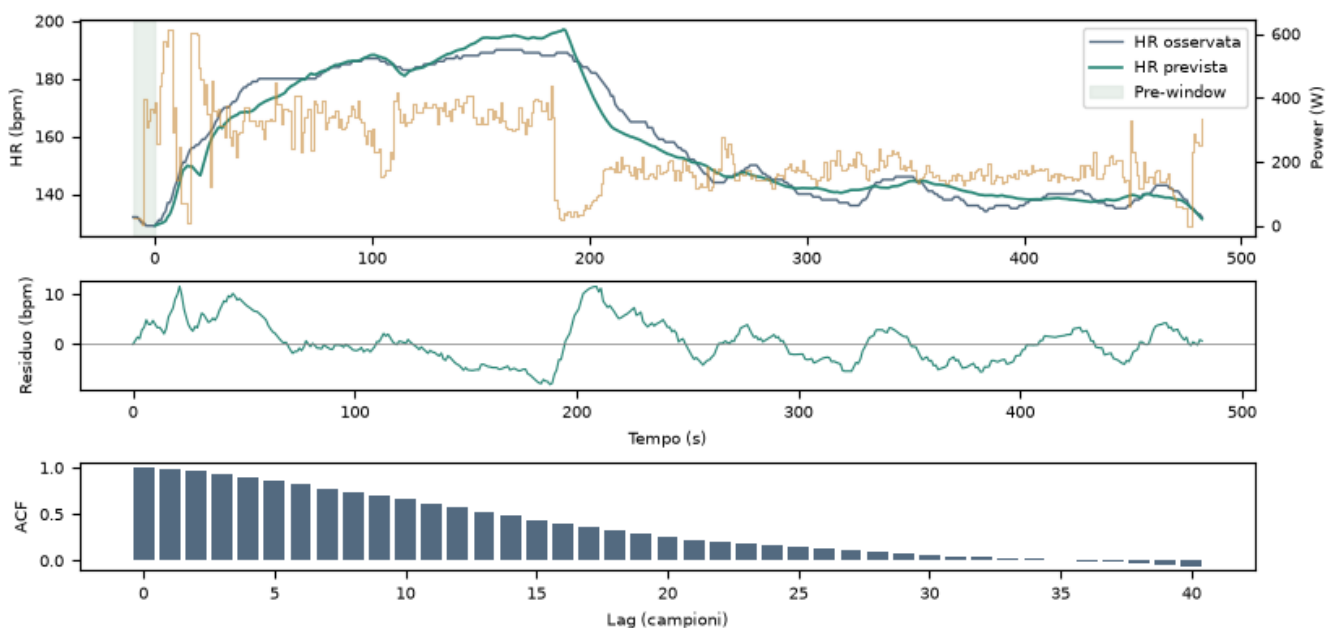
ID: b9e55f3ec4eb05071791d4ee | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=185.1 W; HR0=129 bpm.

HR(0) osservata fissata; B=144.26 bpm stimato; x(0)=-15.255 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
4.1181	3.2657	0.36834	0.96108	1.1159
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.36834	4.1058	0.98648	0.8619	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.34071	0.0041475	0.33258 / 0.34884	1.2173	no	no
L	4.8567	0.34778	4.1751 / 5.5384	7.1607	no	no
tau	49.987	1.0694	47.891 / 52.083	2.1394	no	no
B	144.26	0.28525	143.7 / 144.81	0.19774	no	no

rho=9.5454; Jacobian rank=4/4; cond=454.14; accordo multistart=no.

Correlazioni: {"K_L": -0.26034461508415074, "K_tau": 0.7209440931869677, "K_B": -0.6885013885264804, "L_tau": -0.44168949707787064, "L_B": 0.08916044360649357, "tau_B": -0.49703185901660546}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 8191 / 2.0414e+05

Griglia: 42441 coppie L/tau; SSE multistart originale: 8191; selezione: multistart.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250709173010__laps_019-020.csv

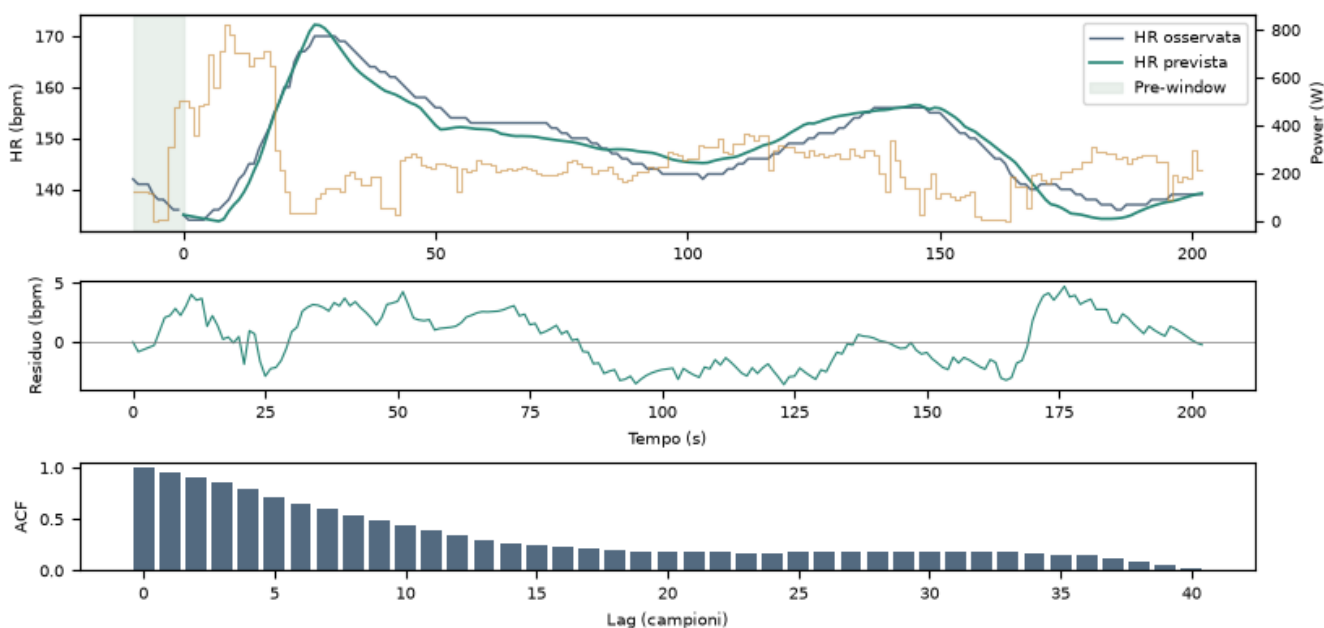
ID: c0e7b9b7347919808a2454d2 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=180 W; HR0=135 bpm.

HR(0) osservata fissata; B=118.61 bpm stimato; x(0)=16.395 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.2412	1.9644	0.25256	0.93673	1.0697
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.25256	2.2324	0.95358	0.72185	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.48673	0.065763	0.35784 / 0.61563	13.511	no	no
L	7.6728	0.23939	7.2036 / 8.142	3.12	no	no
tau	87.339	12.557	62.727 / 111.95	14.377	no	no
B	118.61	4.3106	110.16 / 127.05	3.6344	no	no

rho=2.225; Jacobian rank=4/4; cond=4362.4; accordo multistart=no.

Correlazioni: {"K_L": -0.5981669301025755, "K_tau": 0.9946507609392559, "K_B": -0.998398961133264, "L_tau": -0.6128834370400079, "L_B": 0.5943922164047858, "tau_B": -0.9921797130410067}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1019.6 / 14605

Griglia: 42441 coppie L/tau; SSE multistart originale: 1019.6; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250604173939__laps_005-006.csv

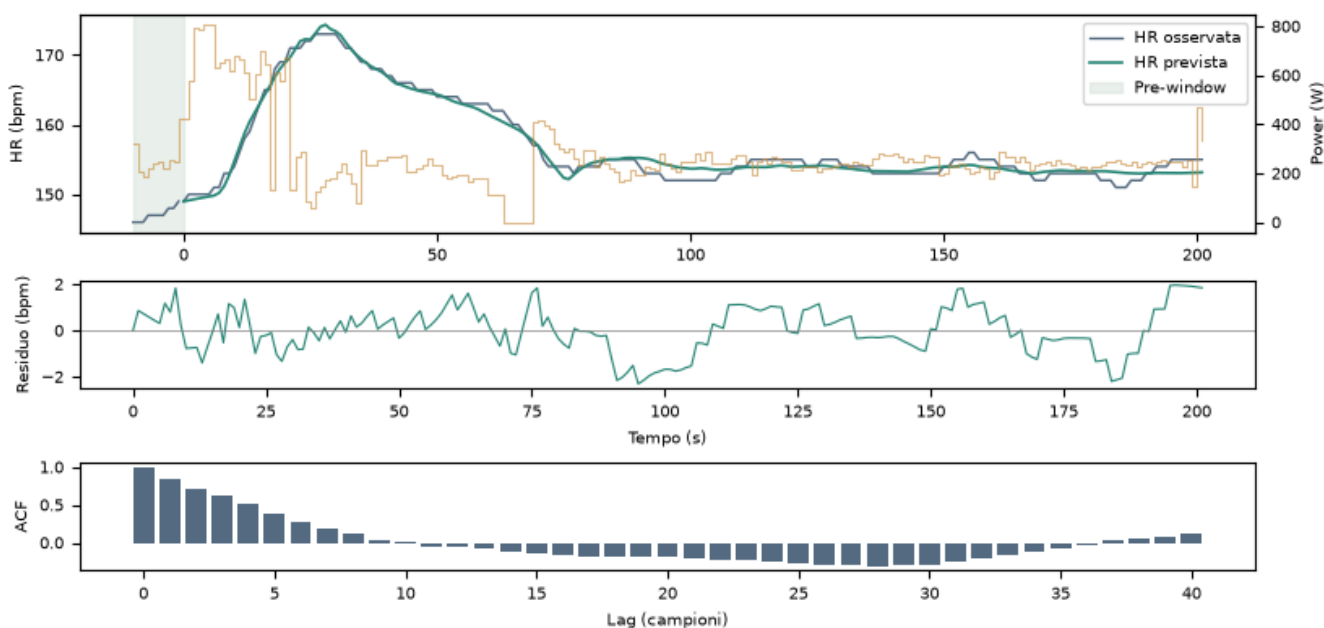
ID: c7ccd097cc8b797f41722582 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=256.2 W; HR0=149 bpm.

HR(0) osservata fissata; B=155.89 bpm stimato; x(0)=-6.8938 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.97676	0.77555	0.017659	0.97518	1.0557
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.017659	0.97903	0.83808	0.39095	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.17217	0.0047474	0.16287 / 0.18148	2.7574	no	no
L	6.5712	0.1596	6.2583 / 6.884	2.4288	no	no
tau	46.966	1.578	43.873 / 50.059	3.3599	no	no
B	155.89	0.09582	155.71 / 156.08	0.061465	no	no

rho=4.1398; Jacobian rank=4/4; cond=1013.6; accordo multistart=no.

Correlazioni: {"K_L": -0.4581912634803313, "K_tau": 0.9432543344960167, "K_B": -0.43074209788070783, "L_tau": -0.4915628886711922, "L_B": 0.09867483753077412, "tau_B": -0.40940307783973073}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 192.72 / 7459.4

Griglia: 42441 coppie L/tau; SSE multistart originale: 192.72; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260501094312__laps_014-015.csv

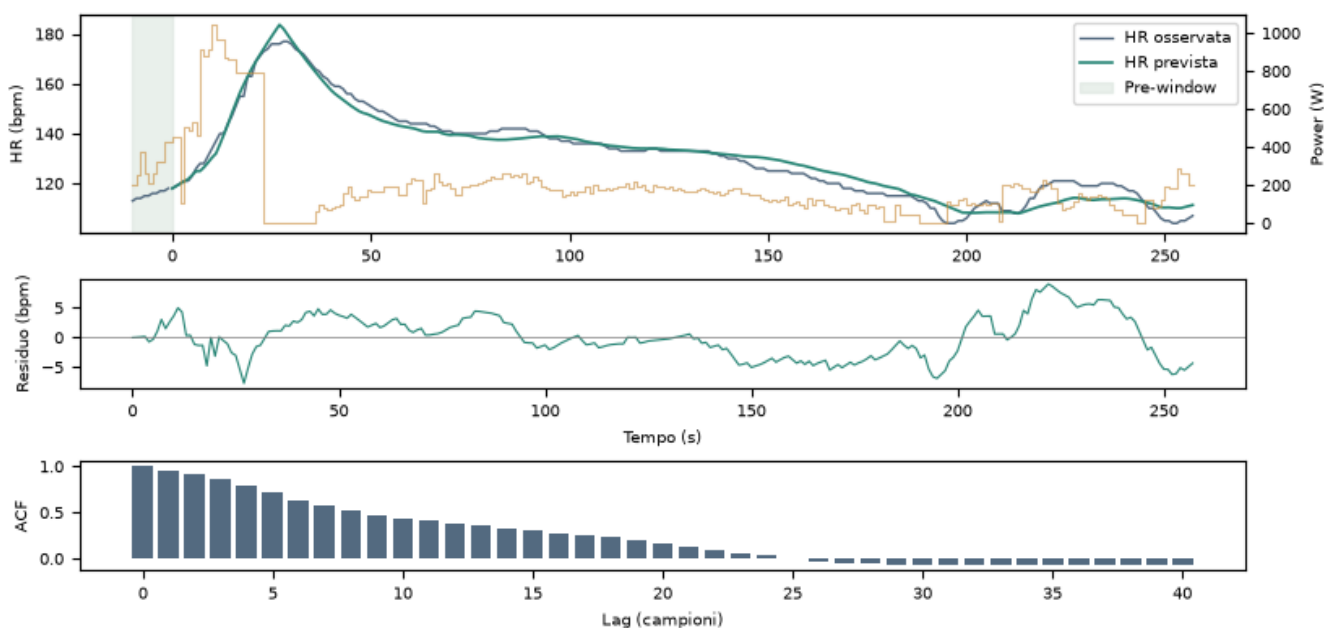
ID: d3beeaabe983fc8c3d8599df | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=307.3 W; HR0=118 bpm.

HR(0) osservata fissata; B=158.92 bpm stimato; x(0)=-40.923 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
3.5126	2.8525	0.10158	0.96112	1.0378
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.10158	3.518	0.95913	0.71538	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.24878	0.0057788	0.23745 / 0.26011	2.3229	no	no
L	4.1807	0.25082	3.6891 / 4.6723	5.9994	no	no
tau	42.024	1.4385	39.205 / 44.843	3.423	no	no
B	158.92	0.65819	157.63 / 160.21	0.41415	no	no

rho=6.0161; Jacobian rank=4/4; cond=900.76; accordo multistart=no.

Correlazioni: {"K_L": -0.3877395437933931, "K_tau": 0.8728802235631169, "K_B": 0.9141484609298814, "L_tau": -0.4027118295420138, "L_B": -0.4318528952626423, "tau_B": 0.758876786257612}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 3183.4 / 81280

Griglia: 42441 coppie L/tau; SSE multistart originale: 3183.4; selezione: multistart.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250709173010__laps_007-008.csv

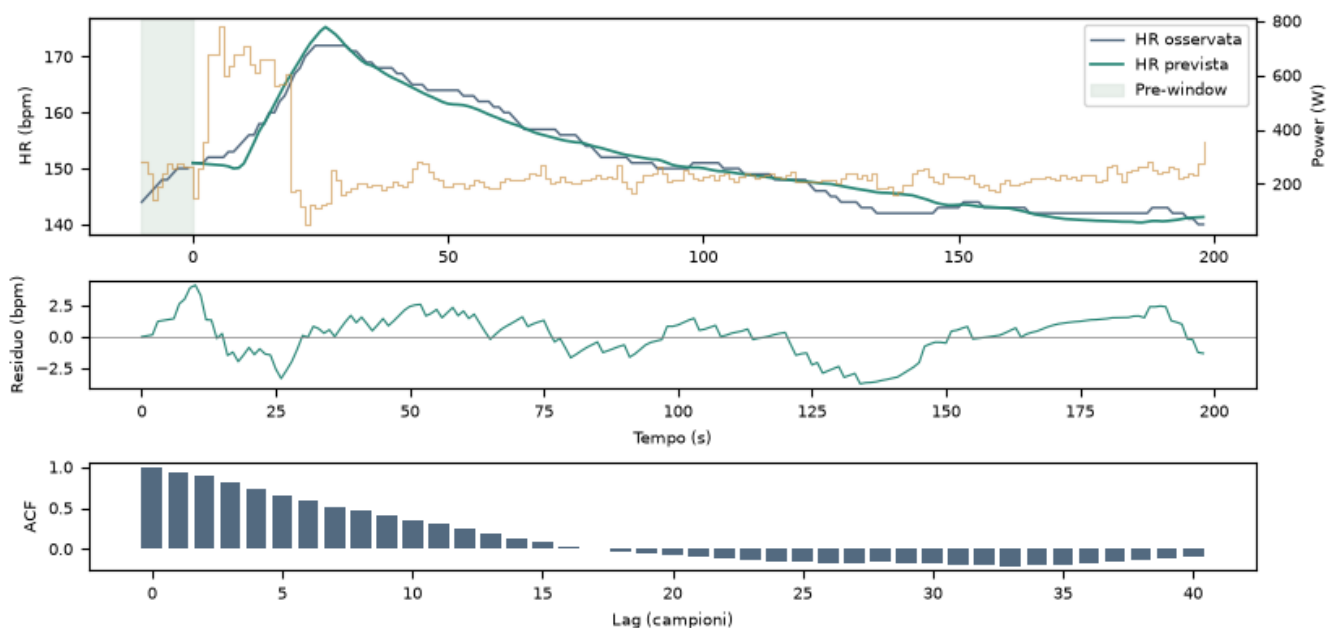
ID: d569267309954c5869e7f9d1 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=243.1 W; HR0=151 bpm.

HR(0) osservata fissata; B=144.43 bpm stimato; x(0)=6.5654 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.5881	1.2704	0.11186	0.9718	1.093
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.11186	1.5882	0.94445	0.65871	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.37911	0.015493	0.34875 / 0.40948	4.0866	no	no
L	6.747	0.23138	6.2935 / 7.2005	3.4294	no	no
tau	85.894	4.4949	77.084 / 94.704	5.2331	no	no
B	144.43	0.47028	143.51 / 145.36	0.3256	no	no

rho=2.2266; Jacobian rank=4/4; cond=984; accordo multistart=no.

Correlazioni: {"K_L": -0.33320915323460976, "K_tau": 0.9523004061682363, "K_B": -0.9076832769390007, "L_tau": -0.35921289384316135, "L_B": 0.2751306192727925, "tau_B": -0.923764506177536}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 501.91 / 15801

Griglia: 42441 coppie L/tau; SSE multistart originale: 501.91; selezione: multistart.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250604173939__laps_015-016.csv

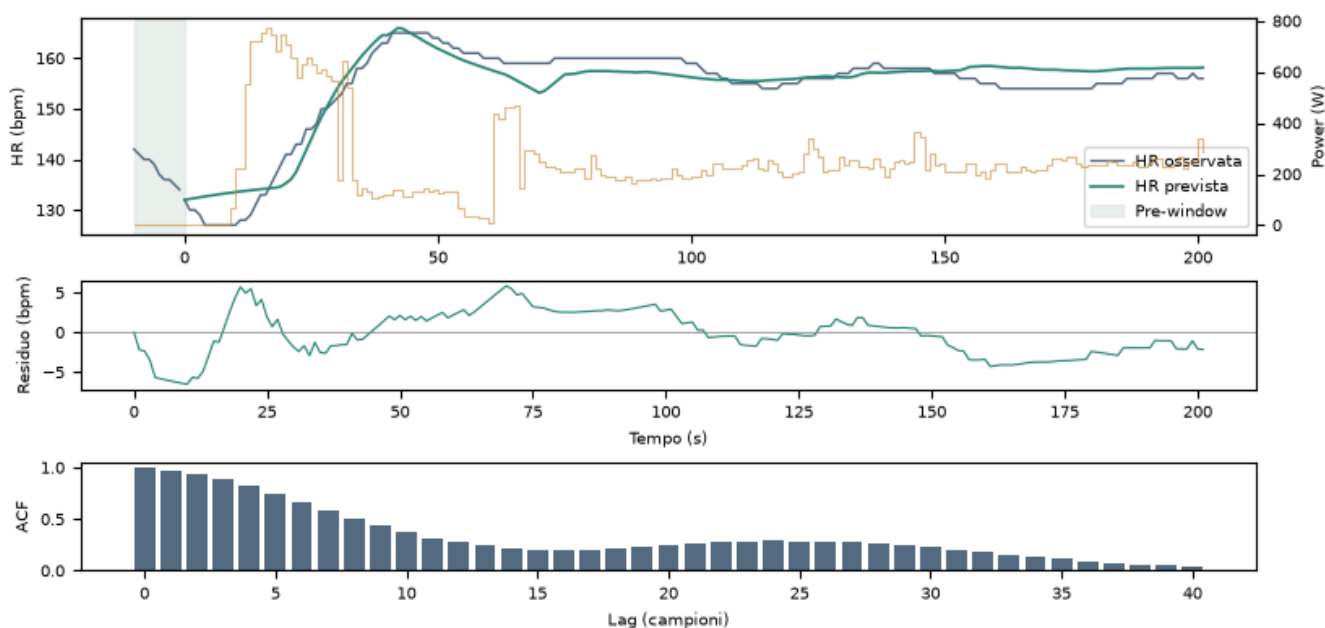
ID: e1f752e5efcb07b2d33a1d4b | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=0 W; HR0=132 bpm.

HR(0) osservata fissata; B=137 bpm stimato; x(0)=-5.0001 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.793	2.3366	-0.16616	0.90353	0.89339
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.16616	2.795	0.96977	0.74627	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.089448	0.0068559	0.076011 / 0.10289	7.6647	no	no
L	9.3293	0.4869	8.375 / 10.284	5.219	no	no
tau	27.38	1.6901	24.067 / 30.693	6.1729	no	no
B	137	1.5164	134.03 / 139.97	1.1068	no	no

rho=7.0004; Jacobian rank=4/4; cond=2382.8; accordo multistart=no.

Correlazioni: {"K_L": -0.551918049570837, "K_tau": 0.811011894003373, "K_B": -0.9863256438397703, "L_tau": -0.6514676241499879, "L_B": 0.5329545232866869, "tau_B": -0.75750265186104}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1575.8 / 5332.3

Griglia: 42441 coppie L/tau; SSE multistart originale: 1575.8; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250709173010__laps_003-004.csv

ID: e62542fef25d6d231ee41d10 | Stato: fitted

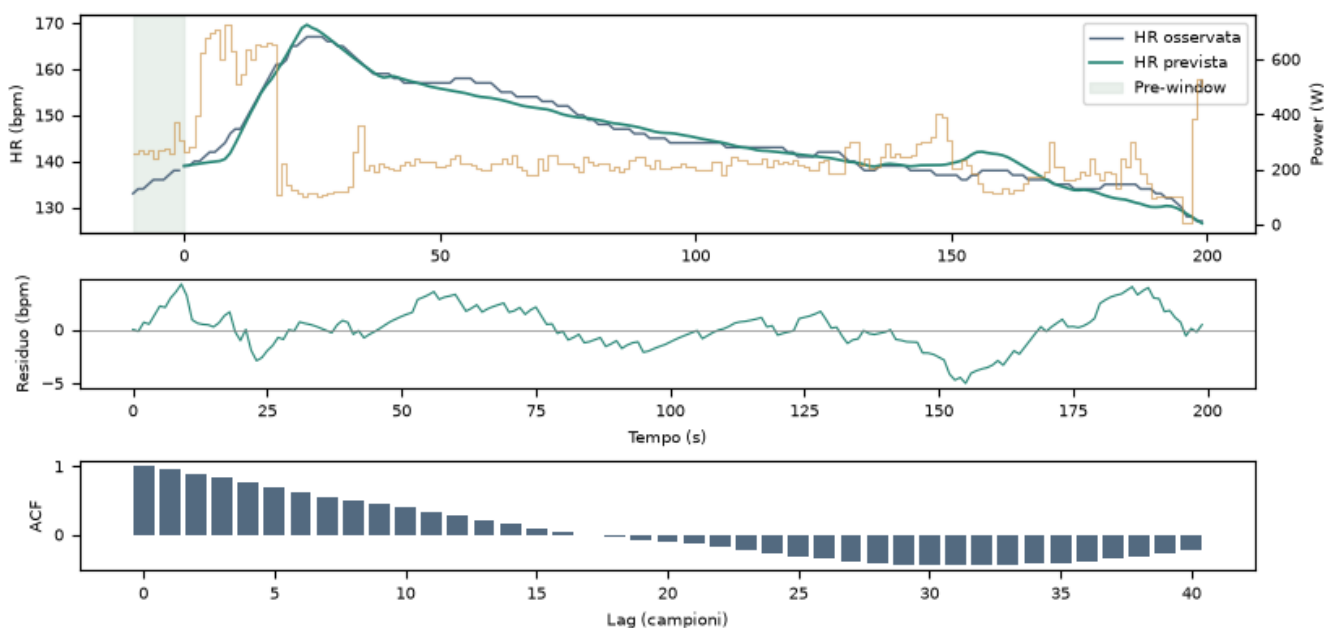
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=278.4 W; HR0=139 bpm.

HR(0) osservata fissata; B=166.95 bpm stimato; $x(0)=-27.947$ bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.8305	1.4106	0.16527	0.96606	1.0764

residual_mean	residual_sd	ACF lag 1	ACF lag 5
0.16527	1.8276	0.95132	0.68912



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.87783	0.083986	0.71321 / 1.0424	9.5675	no	no
L	5.5719	0.22268	5.1355 / 6.0084	3.9964	no	no
tau	168.67	17.852	133.68 / 203.66	10.584	no	no
B	166.95	1.7822	163.45 / 170.44	1.0675	no	no

rho=1.1468; Jacobian rank=4/4; cond=2408.5; accordo multistart=no.

Correlazioni: {"K_L": -0.2573681755619189, "K_tau": 0.9929748796001479, "K_B": 0.9847707979010286, "L_tau": -0.2611685224312714, "L_B": -0.31324376434375895, "tau_B": 0.975012554625103}

Multistart: 16 tentativi, 15 convergenti; SSE min/max: 670.18 / 19581

Griglia: 42441 coppie L/tau; SSE multistart originale: 670.18; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250709173010__laps_023-024.csv

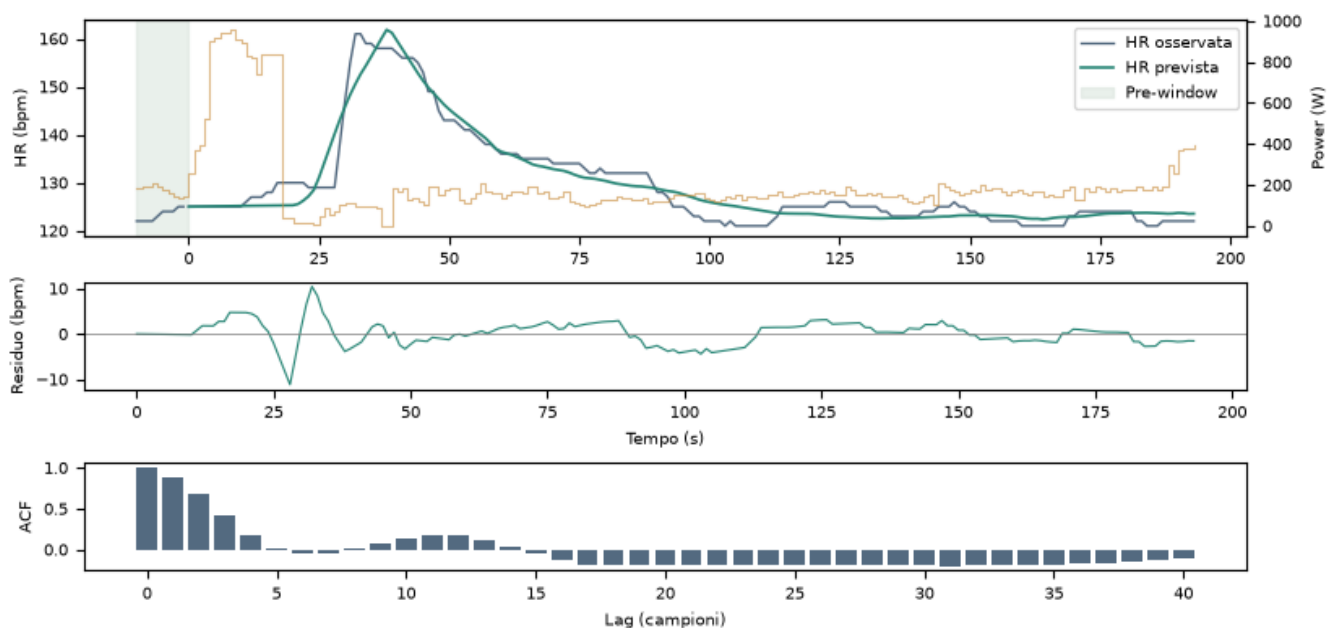
ID: e76e0d896f9264a670919627 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=173.4 W; HR0=125 bpm.

HR(0) osservata fissata; B=125.63 bpm stimato; x(0)=-0.63306 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.5782	1.9504	0.088677	0.93403	0.98727
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.088677	2.5834	0.88859	0.018083	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.13813	0.0050818	0.12817 / 0.14809	3.6789	no	no
L	20.322	0.27314	19.787 / 20.858	1.344	no	no
tau	31.154	1.5713	28.074 / 34.234	5.0438	no	no
B	125.63	0.2804	125.08 / 126.18	0.22319	no	no

rho=5.5427; Jacobian rank=4/4; cond=649.95; accordo multistart=no.

Correlazioni: {"K_L": -0.4286751463554903, "K_tau": 0.8629728410954238, "K_B": -0.6346270390484466, "L_tau": -0.516156865663753, "L_B": 0.24935931024135347, "tau_B": -0.5888713459988971}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1289.6 / 19368

Griglia: 42441 coppie L/tau; SSE multistart originale: 1289.6; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260716153012__laps_006-007.csv

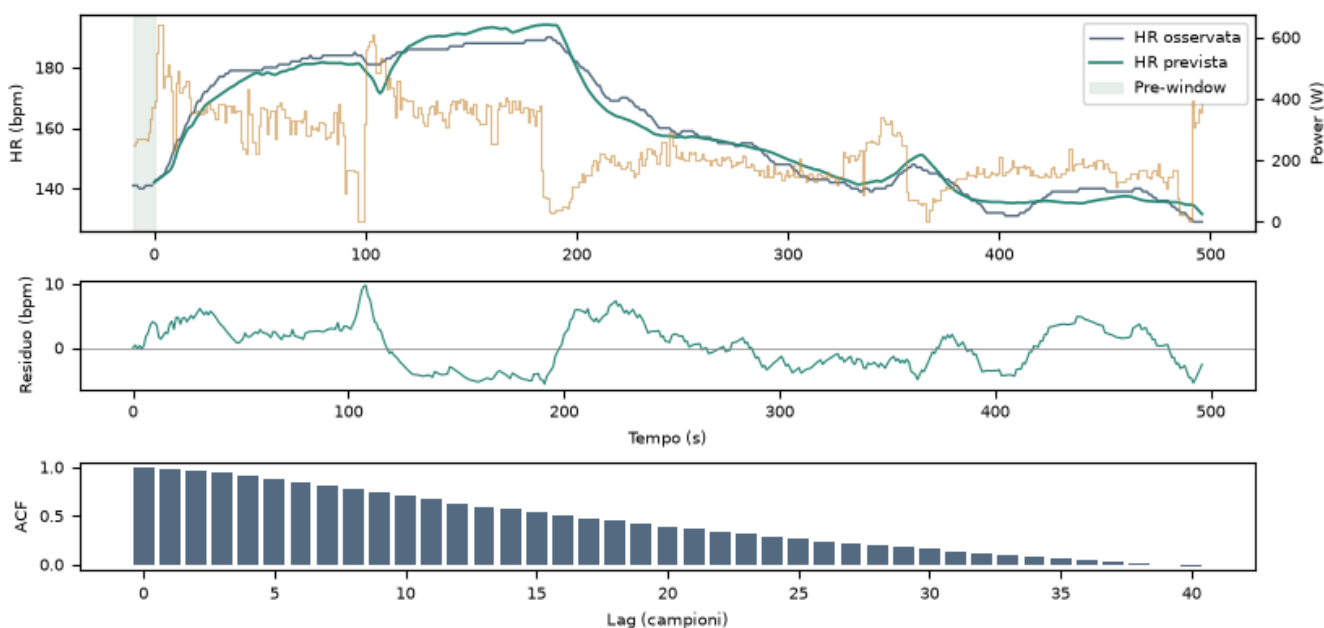
ID: ef94c1727a8f04dac0d1629d | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=285.8 W; HR0=142 bpm.

HR(0) osservata fissata; B=173.4 bpm stimato; x(0)=-31.4 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
3.3954	2.9387	0.26021	0.97216	1.025
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.26021	3.3888	0.98566	0.87591	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.31491	0.0032984	0.30844 / 0.32137	1.0474	no	no
L	7.55	0.33823	6.8871 / 8.213	4.4798	no	no
tau	55.789	1.1016	53.63 / 57.948	1.9745	no	no
B	173.4	0.20904	172.99 / 173.81	0.12055	no	no

rho=8.7553; Jacobian rank=4/4; cond=573.09; accordo multistart=no.

Correlazioni: {"K_L": -0.23803038322610554, "K_tau": 0.7364885335905117, "K_B": 0.553535911565716, "L_tau": -0.350301658886515, "L_B": -0.2890875204303363, "tau_B": 0.337773020303977}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 5729.8 / 2.0394e+05

Griglia: 42441 coppie L/tau; SSE multistart originale: 5729.8; selezione: multistart.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260501094312__laps_002-003.csv

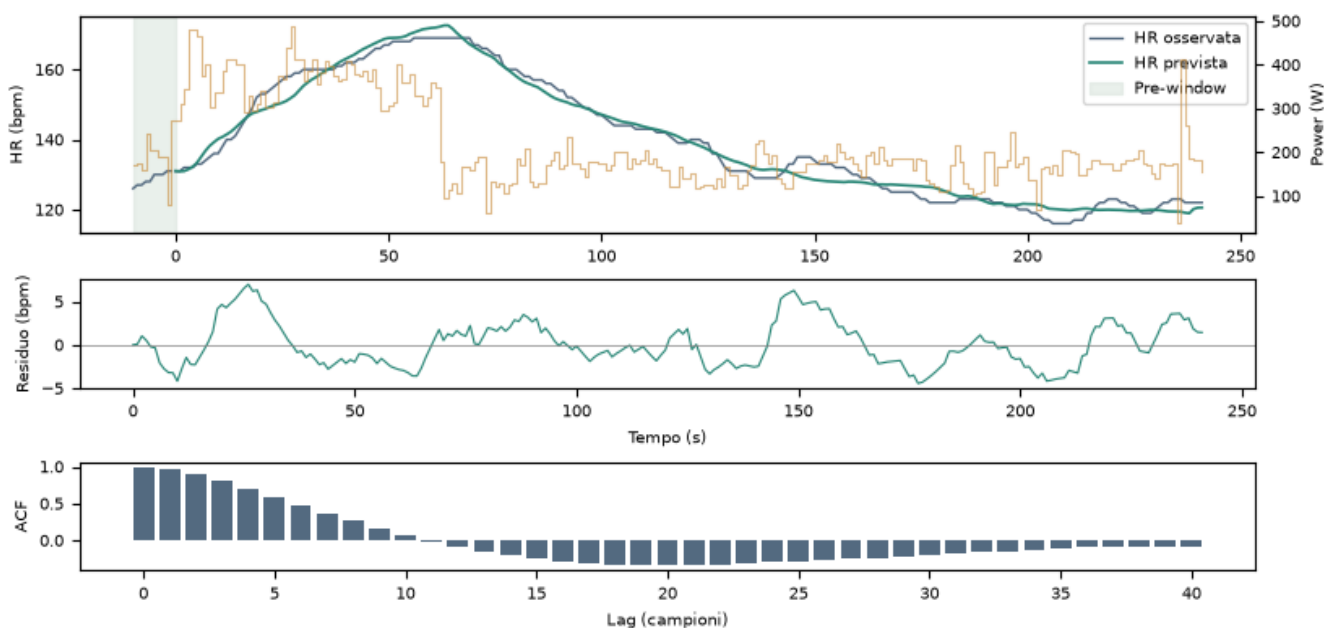
ID: fa162b94f965b809d369523a | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=188.7 W; HR0=131 bpm.

HR(0) osservata fissata; B=124.84 bpm stimato; $x(0)$ =6.1586 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.5902	2.1209	0.055166	0.97689	1.0129
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.055166	2.595	0.95691	0.59232	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.43765	0.011526	0.41506 / 0.46024	2.6336	no	no
L	1.7117	0.37969	0.96746 / 2.4558	22.183	no	no
tau	68.429	2.5907	63.351 / 73.507	3.7859	no	no
B	124.84	0.55346	123.76 / 125.93	0.44333	no	no

rho=3.4969; Jacobian rank=4/4; cond=746.7; accordo multistart=no.

Correlazioni: {"K_L": -0.5210361757320442, "K_tau": 0.9367038435963179, "K_B": -0.9079010389177332, "L_tau": -0.5936506434754203, "L_B": 0.42799293275483935, "tau_B": -0.8866259229283919}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1623.7 / 69875

Griglia: 42441 coppie L/tau; SSE multistart originale: 1623.7; selezione: multistart.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260716153012__laps_010-011.csv

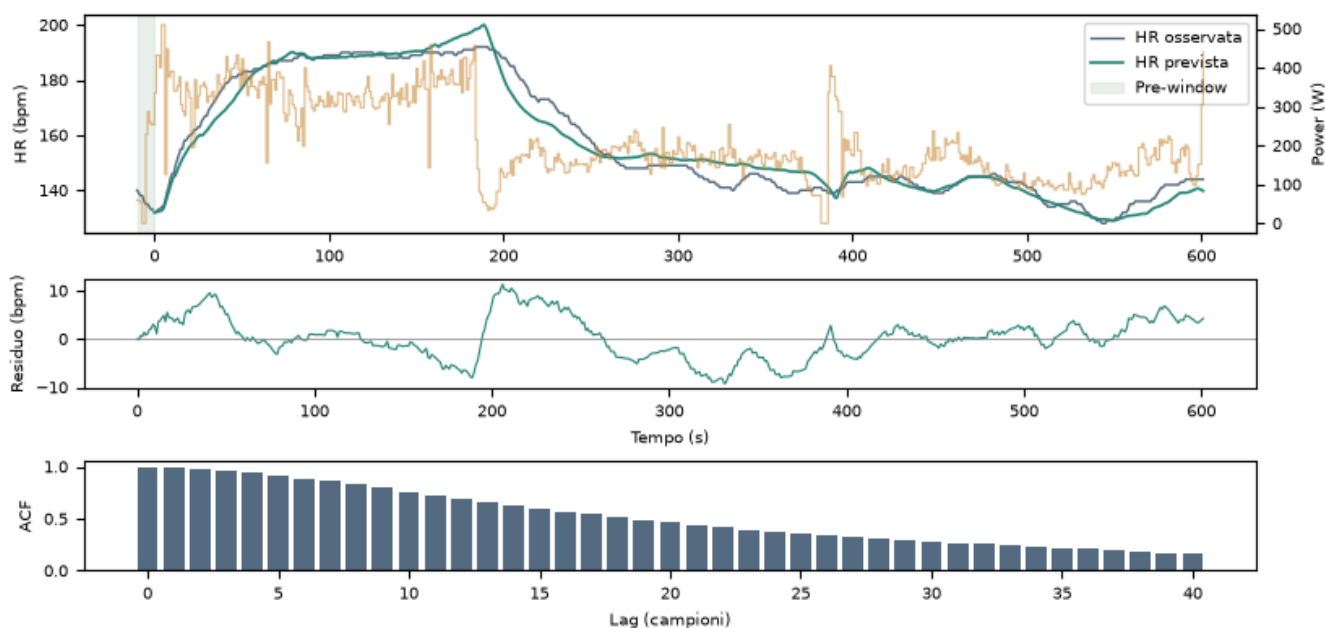
ID: fe481369e80e480c9bff4c05 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=145.5 W; HR0=132 bpm.

HR(0) osservata fissata; B=138.84 bpm stimato; x(0)=-6.8438 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
4.2905	3.3541	0.2075	0.95873	1.1082
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.2075	4.2891	0.98944	0.91426	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.30078	0.0031259	0.29466 / 0.30691	1.0393	no	no
L	4.6595	0.39251	3.8902 / 5.4289	8.4237	no	no
tau	42.785	0.99205	40.841 / 44.73	2.3187	no	no
B	138.84	0.28012	138.29 / 139.39	0.20175	no	no

rho=13.938; Jacobian rank=4/4; cond=553.48; accordo multistart=no.

Correlazioni: {"K_L": -0.2790542399187954, "K_tau": 0.633963331876717, "K_B": -0.7457535478719699, "L_tau": -0.514609931530749, "L_B": 0.16936955075032306, "tau_B": -0.49426304001227095}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 11082 / 2.639e+05

Griglia: 42441 coppie L/tau; SSE multistart originale: 11082; selezione: multistart.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Configurazione completa dell'esperimento

Configurazione, split, sorgenti, hash e ambiente della run. I CSV originali restano conservati nel blob store. Manifest integrale e risultati sono inclusi nell'export tabellare ZIP.

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    {
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    {
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  ],
  "equation": "tau dx/dt + x = K (P(t-L)-P0); HRhat = B+x",
  "source": "PID richiesto nel protocollo utente; nessuna replica bibliografica rivendicata",
  "initial_conditions": "P0=mean(Ppre); HR0=HR(0); x(0)=HR0-B; B estimated; P(t<0)=P0",
  "input_reconstruction": "zero-order hold, integrazione esatta; ritardo continuo in secondi",
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  "delay_domain": "0 <= L <= min(Lmax,T); T=t[-1]-t[0]; Lmax=null significa T; Lmax=0 fissa L=0",
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    "labels": [
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    "seed": 42,
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